Lecture Notes on Holography

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Preface

Holography, also known as the anti-de Sitter/conformal field theory (AdS/CFT) correspondence or gauge/gravity duality, is the conjecture that certain quantum field theories (QFTs) in d dimensions are equivalent to certain theories of gravity in $(d+1)$-dimensions. Pairs of theories related by holography are said to be *holographically dual* to one another. Observables in a quantum field theory can be mapped to observables in its holographically dual gravitational theory, and vice versa. This is a great advantage when one of the theories is much simpler than the other; we can use the simpler theory to make predictions for the more complicated one.

Holography has been a major topic of research in theoretical high energy physics for the past three decades, for at least two reasons:

- Quantum field theory is much better understood than quantum gravity. A theory of quantum gravity with a holographic dual QFT can be studied through the QFT.
- QFTs with many degrees of freedom and large coupling constants are typically holographically dual to theories of classical gravity (general relativity plus some matter fields). Large coupling constants mean we cannot use perturbation theory, the usual way to calculate things in QFT. On the other hand, classical gravity is relatively easy to compute in. Thus, holography can be a useful to compute observables in strongly coupled QFTs.

In these lectures we will focus on the latter approach, using gravitational calculations to make predictions for strongly coupled QFT. We will mostly focus on the canonical example of the AdS/CFT correspondence, in which the QFT is a particular four-dimensional gauge theory called $\mathcal{N}=4$ supersymmetric Yang–Mills (SYM) theory.

1 Aspects of quantum field theory and gravity

We will begin by discussing some aspects of quantum field theory (QFT) and general relativity. Some of the things we discuss will be review of topics that are standard in QFT courses, and the intention is to highlight things that will be important for holography, as well as to establish some notation that will be used throughout this course. We may also cover some new concepts. Throughout this course we use natural units in which the reduced Planck constant, the speed of light, and Boltzmann's constant all equal one,

$$\hbar = c = k_{\mathrm{B}} = 1\,. \tag{1.1}$$

We always use the mostly-plus metric signature, i.e. we take the Minkowski metric in d spacetime dimensions to be

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + \mathrm{d}x_1^2 + \cdots + \mathrm{d}x_{d-1}^2\,. \tag{1.2}$$

1.1 Path integrals in quantum field theory

Generating functionals. A central object in the path integral formulation is the partition function Z. If the QFT has fields ϕ and action $S[\phi]$, the partition function is

$$Z = \int \mathcal{D}\phi\, e^{iS[\phi]}\,. \tag{1.3}$$

Suppose a QFT contains some local operators $\{\mathcal{O}_a(x)\}$. Key observables in quantum field theory are the correlation functions of these operators,

$$\langle \mathcal{O}_{a_1}(x_1)\mathcal{O}_{a_2}(x_2)\cdots\mathcal{O}_{a_n}(x_n)\rangle \equiv \frac{1}{Z}\int \mathcal{D}\phi\, \mathcal{O}_{a_1}(x_1)\mathcal{O}_{a_2}(x_2)\cdots\mathcal{O}_{a_n}(x_n)e^{iS[\phi]}\,. \tag{1.4}$$

These correlation functions can be encoded in the generating functional $Z[J]$, which depends on *sources* $J^a(x)$, one for each operator. It is defined by

$$Z[J] = \int \mathcal{D}\phi\, \exp\left(iS + i\int \mathrm{d}^d x\, J^a(x)\mathcal{O}_a(x)\right), \tag{1.5}$$

where d denotes the dimension of spacetime and the a index is summed over. One can take functional derivatives of the generating functional to obtain correlation functions in the presence of sources, denoted $\langle\cdot\rangle_J$,

$$\begin{aligned} \langle \mathcal{O}_a(x)\rangle_J &= -i\frac{1}{Z}\frac{\delta Z[J]}{\delta J^a(x)}\,, \\ \langle \mathcal{O}_a(x)\mathcal{O}_b(y)\rangle_J &= (-i)^2\frac{1}{Z}\frac{\delta^2 Z[J]}{\delta J^a(x)\delta J^b(y)}\,, \end{aligned} \tag{1.6}$$

and so on. The correlation functions in the original QFT are then obtained by setting $J = 0$. It is common to define the generating functional of connected correlation functions $W[J] = -i\log Z[J]$. The connected correlation functions in the presence of sources $\langle\cdot\rangle_{\mathrm{c},J}$ are then given by functional derivatives of W,

$$\langle \mathcal{O}_{a_1}(x_1)\cdots\mathcal{O}_{a_n}(x_n)\rangle_{\mathrm{c},J} = (-i)^{n-1}\frac{\delta^n W}{\delta J^{a_1}(x_1)\cdots\delta J^{a_n}(x_n)}\,. \tag{1.7}$$

Classical limit. The classical limit arises when we consider QFTs for which the action is typically large. Then the factor $e^{iS[\phi]}$ tends to oscillate rapidly as a functional of the field configuration, and these fluctuations tend to average to zero in the path integral. The exception is for field configurations for which the action is stationary, i.e.

$$\frac{\delta S[\phi]}{\delta\phi} = 0\,. \tag{1.8}$$

By definition, close to such a configuration the action is approximately independent of ϕ, so $e^{iS[\phi]}$ does not oscillate so rapidly and we can have a non-zero contribution to the path integral. Equation (1.8) is the Euler–Lagrange equation, i.e. the classical equations of motion, so our statement is that solutions to the classical equations of motion dominate the path integral. Consequently, in the classical limit one can make a saddle-point approximation to the path integral,

$$Z \approx e^{iS^\star}\,, \tag{1.9}$$

where $S^\star$ denotes the *on-shell action*, which means $S[\phi]$ evaluated on a solution to the classical equations of motion (1.8).

Wick rotation to Euclidean signature and thermal field theory. The oscillating $e^{iS[\phi]}$ factor in the path integral makes it somewhat ill-defined. For this reason, it is common to perform a *Wick rotation*, which works as follows. The action is the integral of the Lagrangian L over time t,

$$S[\phi] = \int_{-\infty}^{\infty} \mathrm{d}t \, L(\phi; t) \,. \tag{1.10}$$

We can think of this as a contour integral in the complex t plane, with the contour running along the real axis. We then rotate the contour so that it instead runs along the imaginary axis by definining the *imaginary time* $\tau = it$ and integrating over real τ. This defines the Euclidean action which we denote I,

$$I[\phi] = -\int_{-\infty}^{\infty} \mathrm{d}\tau \, L(\phi, -i\tau) = -iS[\phi] \,. \tag{1.11}$$

We call I the Euclidean action because for real τ the d-dimensional Minkowski metric becomes that of d-dimensional Euclidean space,

$$-\,\mathrm{d}t^2 + \mathrm{d}\vec{x}^{\,2} = \mathrm{d}\tau^2 + \mathrm{d}\vec{x}^{\,2} \,, \tag{1.12}$$

so going to imaginary time changes the signature of spacetime from Lorentzian to Euclidean. In Euclidean signature, the partition function becomes

$$Z = \int \mathcal{D}\phi \, e^{-I[\phi]} \,. \tag{1.13}$$

This has better convergence properties than the Lorentzian path integral because of the decaying, rather than oscillating, exponential.

Working in Euclidean signature also provides a convenient way to study thermodynamics. The thermodynamic partition function of a quantum system with Hamiltonian H at inverse temperature β is

$$Z(\beta) = \mathrm{tr}\, e^{-\beta H} \,. \tag{1.14}$$

That this is indeed the thermodynamic partition function can be seen by evaluating the trace in the basis of energy eigenstates $|n\rangle$, with energies E_n,

$$Z(\beta) = \sum_n \langle n| e^{-\beta H} |n\rangle = \sum_n e^{\beta E_n} \,. \tag{1.15}$$

Since the trace is independent of basis, in a QFT we could also choose to evaluate the trace in the basis of states specified by field configurations at $\tau = 0$. This leads to a path integral expression for the thermodynamic partition function

$$Z(\beta) = \int_{\phi(0)=\pm\phi(\beta)} \mathcal{D}\phi \, e^{-I[\phi]} \,, \tag{1.16}$$

where the imaginary time runs only over the range $0 \leq \tau \leq \beta$, and the notation indicates that we should integrate over all field configurations such that $\phi(\tau = 0, \mathbf{x}) = \pm\phi(\tau = \beta, \mathbf{x})$, where one should choose the plus sign for bosons and the minus sign for fermions. In other words:

> The thermodynamic partition function is obtained by the path integral over all field configurations which are (anti)periodic in imaginary time, with period equal to the inverse temperature β.

1.2 Large-N gauge theory

Yang–Mills theory with gauge group SU(N) plays an important role in the standard model, and also in holography. Some simplifications arise in this theory in the limit $N \gg 1$. The action for the theory is

$$S = -\frac{1}{2g_{\mathrm{YM}}^2} \int \mathrm{d}^4x \, \mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) \,, \tag{1.17}$$

where g_{YM} is the coupling constant and $F_{\mu\nu}$ is the gauge field strength. If we want to take the $N \to \infty$ limit we should also rescale g_{YM}. The reason is that it is the quantity $\lambda = g_{\mathrm{YM}}^2 N$, called the *'t Hooft coupling*, that really controls the size of perturbative corrections at large N. If we send $N \to \infty$, we should also send $g_{\mathrm{YM}}^2 \to 0$ in such a way that λ remains fixed.

To see why the 't Hooft coupling is important in the large-N limit, consider the one-loop beta function for g_{YM}^2,

$$\beta(g_{\mathrm{YM}}^2) \equiv \mu \frac{\mathrm{d}g_{\mathrm{YM}}^2}{\mathrm{d}\mu} = -\frac{11 g_{\mathrm{YM}}^4 N}{24\pi^2} \,. \tag{1.18}$$

The right-hand side shows that it does not make sense to keep g_{YM}^2 fixed when sending $N \to \infty$; if you keep g_{YM}^2 fixed at some scale $\mu = \mu_0$, the beta function shows that at other scales g_{YM}^2 will be shifted by a factor proportional to N. On the other hand, multiplying both sides of equation (1.18) by N we find that the one-loop beta function for the 't Hooft coupling is

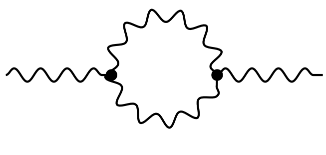
$$\beta(\lambda) = -\frac{11\lambda^2}{24\pi^2} \,. \tag{1.19}$$

So it does make sense to keep λ fixed in the large-N limit.

In more detail, let's consider how an example Feynman diagram behaves at large N. Recall that with the gauge field normalised as in equation (1.17), the gluon propagator is proportional to the square of the Yang–Mills coupling,

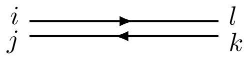
$(A_\mu)^i{}_j$  $(A_\nu)^k{}_l \quad \propto g_{\mathrm{YM}}^2 \left(\delta^i_l \delta^k_j - \frac{1}{N} \delta^i_j \delta^k_l \right)$ (1.20)

where i, j etc are the gauge indices — each A_μ is an $N \times N$ traceless hermitian matrix. Since we are working at large N, we will drop the second term in the brackets, since it goes to zero when $N \to \infty$. With the gauge field normalised as in equation (1.17), the interaction vertices are proportional to $1/g_{\mathrm{YM}}^2$. Now consider the following one-loop self-energy diagram below, which at large N scales with g_{YM} and N as follows:

 $\sim g_{\mathrm{YM}}^4 N$ (1.21)

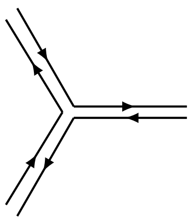
The diagram goes as $g_{\mathrm{YM}}^4 = g_{\mathrm{YM}}^{2(4-2)}$ because there are four propagators and two vertices. It is proportional to N because the closed loop involves a trace over the gauge indices, which run from 1 to N. Comparing equations (1.20) and (1.21), we see that the one-loop diagram is smaller than the tree level diagram by a factor of $\lambda = g_{\mathrm{YM}}^2 N$.

To keep track of the traces in the diagrammatics, in large-N Yang–Mills theory it is common to use ‘t Hooft’s double line notation. In this notation, the large-N limit of the gauge field propagator in equation (1.20) is denoted by two lines, one representing δ^i_l and another representing δ^k_j, with arrows pointing from the upper to the lower indices. The propagator then looks like this

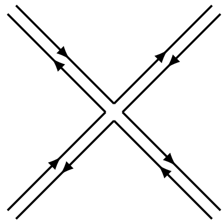


$$\sim \frac{\lambda}{N} \tag{1.22}$$

where we have indicated that the propagator is proportional to $g^2_{\mathrm{YM}} = \lambda/N$. In this notation the interaction vertices, which both scale as N/λ, look like

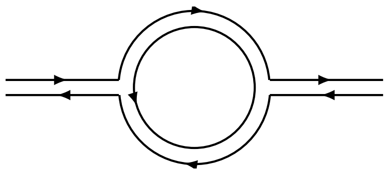


$$\sim \frac{N}{\lambda}$$



$$\sim \frac{N}{\lambda} \tag{1.23}$$

For example, in the double line notation the one-loop diagram in equation (1.21) looks like this

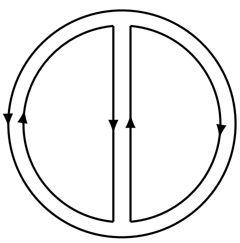


$$\sim \left(\frac{\lambda}{N}\right)^4 \left(\frac{N}{\lambda}\right)^2 N = \frac{\lambda^2}{N} \tag{1.24}$$

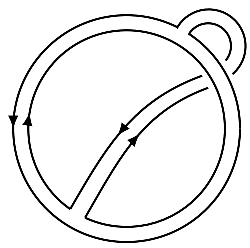
In this notation, the factor of N coming from the gauge index trace is represented by the (single line) closed loop at the centre of the diagram. More complicated diagrams can have more of these index loops. In general, a diagram with p propagators, v vertices, and ℓ index loops scales with with λ and N as

$$\text{diagram} \sim \left(\frac{\lambda}{N}\right)^p \left(\frac{N}{\lambda}\right)^v N^\ell = N^{v+\ell-p}\,\lambda^{p-v}\,. \tag{1.25}$$

As further examples, let’s look at two vacuum diagrams, both at occurring at one loop, applying the rule above to see how they scale with λ and N:



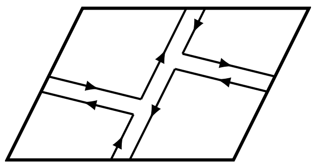
$$\sim \lambda N^2$$



$$\sim \lambda \tag{1.26}$$

Both diagrams have three propagators and two vertices, which combine to give a factor of λ/N. The momentum structure of the two diagrams is the same, but the gauge indices are contracted in different ways, leading to different scaling with N. The diagram on the left has three closed loops, giving a factor of N^3 so that in total the diagram scales as λN^2. The diagram on the right only has one closed loop, so scales instead as λN. Consequently, the diagram on the left of equation (1.26) provides the dominant one-loop contribution to the vacuum at large N

The diagram on the left of equation (1.26) can be drawn on a plane without any lines crossing, so is called a planar diagram. The diagram on the right cannot, so it a non-planar diagram. Non-planar diagrams *can* be drawn without lines crossing on higher genus Riemann surfaces. The diagram on the right of equation (1.26) be drawn on a torus without any lines crossing. Here is how the diagram looks on a torus, drawn as a parallelogram with opposite sides identified:



Notice that the diagram tiles the torus, i.e. every part of the torus is part of a region bounded by the propagators (once you shrink the gap between two lines forming the propagator to zero). In fact, in this example there is just a single such region. There is a topological number χ called the Euler characteristic that can be associated to every Riemann surface. Given a tiling of the surface, the Euler characteristic is given by

$$\chi = (\text{number of vertices}) - (\text{number of edges}) + (\text{number of faces})\,. \tag{1.27}$$

It is independent of how the surface is tiled. In the tiling provided by a Feynman diagram, the edges of the tiling are the propagators while the faces are formed by the index loops. The vertices are just the vertices of the diagram. Hence we have $\chi = p + \ell - v$. This is exactly the exponent of N appearing in equation (1.25), so we can rewrite the rule for the large-N scaling of a Feynman diagram as

$$\text{diagram} \sim N^{\chi} \lambda^{p-v} = N^{2-2\mathrm{g}} \lambda^{p-v}\,, \tag{1.28}$$

where on the right-hand side we used the relation $\chi = 2 - 2\mathrm{g}$, where g is the genus (number of handles) of the Riemann surface. To make this formula work for planar diagrams we should add the point at infinity to the plane, turning it into the Riemann sphere which has Euler characteristic $\chi = 2$, matching for example the N^2 scaling of the planar diagram on the left of equation (1.25).

1.3 Conformal symmetry

Conformal transformations are symmetry transformations that leave angles between vectors unchanged. We can phrase this in terms of the metric as follows. Under a general coordinate transformation $x \to x'$, the metric $g_{\mu\nu}$ transforms as $g_{\mu\nu}(x) \to g'_{\mu\nu}(x') = (\partial x^\rho / \partial x'^\mu)(\partial x^\sigma / \partial x'^\nu) g_{\rho\sigma}(x)$. Conformal transformations are those coordinate transformations for which

$$g'_{\mu\nu} = \Omega(x)^2 g_{\mu\nu}\,, \tag{1.29}$$

for some function $\Omega(x)$. In other words, conformal transformations only change the metric by a scale factor.

Two examples of conformal transformations are translations and rotations, which leave the Minkowski metric invariant, i.e. they have $\Omega(x) = 1$. An example of a conformal transformation with non-trivial Ω is a constant rescaling of the coordinates $x \to \alpha x$, with $\Omega(x) = \alpha^{-1}$. Finally, there are so-called special conformal transformations, specified by a constant vector b^μ, under which

$$x^\mu \to \frac{x^\mu + b^\mu x^2}{1 + 2b_\nu x^\nu + b^2 x^2}\,, \tag{1.30}$$

with $\Omega(x) = 1 + 2b_\nu x^\nu + b^2 x^2$. This is the complete set of conformal transformations. In d spacetime dimensions they form the group $\mathrm{SO}(d, 2)$.

A conformal field theory (CFT) is a quantum field theory that is invariant under conformal transformations. Classically, a theory is conformal if it does not have any dimensionful parameters; any such parameters would be incompatible with invariance under scale symmetry $x \to \alpha x$. For example, Yang–Mills theory in four dimensions is classically conformal. However, quantization of a field theory typically introduces a dimensionful quantity, the renormalisation scale, which breaks conformal symmetry. Dependence on the renormalisation group scale is encoded in the beta functions for couplings.

For instance, in $\mathrm{SU}(N)$ Yang–Mills theory minimally coupled to n_s massless scalar fields in the representation R_s of $\mathrm{SU}(N)$ and n_f Weyl fermions in the representation R_f, the one-loop beta function for the gauge coupling is

$$\beta(g_{\mathrm{YM}}) = -\left[\frac{11}{3}N - \frac{n_s}{6}T(R_s) - \frac{2n_f}{3}T(R_f)\right]\frac{g_{\mathrm{YM}}^3}{16\pi^2}\,, \tag{1.31}$$

where $T(R)$ is the Dynkin index of the representation R.[1] Even if you choose N, n_s, and n_f such that the one-loop beta function vanishes, the two-loop contribution to the beta function will be non-zero. In general, QFTs are conformal only at fixed points, where by definition all of the beta functions vanish.

An example of a QFT which *is* conformal is called four-dimensional $\mathcal{N} = 4$ supersymmetric Yang–Mills (SYM) theory. This is a supersymmetric version of Yang–Mills theory, which in addition to the gauge field contains six scalar fields ϕ^i and four Weyl fermions ψ_a, all transforming in the adjoint representation with interactions dictated by conformal symmetry. We will always take the gauge group to be $\mathrm{SU}(N)$.

The action for $\mathcal{N} = 4$ SYM is

$$\begin{aligned} S = \int \mathrm{d}^4 x\, \mathrm{tr}\Big(&-\frac{1}{2g_{\mathrm{YM}}^2} F_{\mu\nu}F^{\mu\nu} - D_\mu \phi^i D^\mu \phi^i + \frac{g_{\mathrm{YM}}^2}{2}[\phi^i, \phi^j][\phi^i, \phi^j] \\ &+ i\psi_a \bar{\sigma}^\mu D_\mu \psi_a + \frac{g_{\mathrm{YM}}}{2} C^i_{ab} \psi_a [\phi^i, \psi_b] + \frac{g_{\mathrm{YM}}}{2} \bar{C}^i_{ab} \bar{\psi}_a [\phi^i, \bar{\psi}_b] \Big)\,, \end{aligned} \tag{1.32}$$

where D_μ is the covariant derivative, $\bar{\sigma^\mu} = (\mathbb{1}, -\vec{\sigma})$ is a four-vector of Pauli matrices, $\bar{\psi}_a$ is the conjugate transpose of ψ_a. We employ the notation that two Weyl spinors written next to each other have their indices contracted with the two-dimensional Levi–Civita symbol, $\psi_a \psi_b \equiv \epsilon^{\alpha\beta} \psi_{a,\alpha} \psi_{b,\beta}$. The quantities C^i_{ab} and $\bar{C}^i_{ab}$ appearing in the scalar–fermion interaction terms are numbers fixed by some group theory. The theory has an

[1] The fundamental representation $\mathbf{N}$ has $T(\mathbf{N}) = 1/2$. The adjoint representation adj has $T(\mathrm{adj}) = N$.

SO(6) symmetry which rotates the six scalar fields, and under which the fermions also transform. The C^i_{ab} and $\bar{C}^i_{ab}$ are Clebsch–Gordan coefficients relating the six- and four-dimensional representations of SO(6). We won't give these coefficients explicitly since the only calculations we will do directly in $\mathcal{N} = 4$ SYM will be for the free theory with $g_{\mathrm{YM}} = 0$.

Plugging $n_s = 6$ and $n_f = 4$ into equation (1.31) and using that for the adjoint representation $T(R) = N$, we see that the one-loop beta function vanishes for $\mathcal{N} = 4$ SYM. The magic of the theory is that all of the higher-loop corrections vanish too, so that the beta function is zero to all orders and therefore $\mathcal{N} = 4$ SYM is a genuine conformal field theory.

We characterise operators in a CFT by how they transform under conformal transformations. An important class of operators are *scalar primary operators.* A scalar primary operator $\mathcal{O}(x)$ transforms under conformal transformations as

$$\mathcal{O}(x) \to \Omega(x')^{\Delta}\mathcal{O}(x')\,, \tag{1.33}$$

for some real number Δ called the *scaling dimension* of the operator. In a free theory the scaling dimension is just the engineering dimension of the operator. In an interacting theory the scaling dimension can receive quantum corrections. A scalar primary which exists in every CFT is the identity operator $\mathbb{1}$, which always has $\Delta = 0$. Apart from the identity operator, the scaling dimensions of scalar primaries in a d-dimensional CFT obey the *unitarity bound*

$$\Delta \geq \frac{d-2}{2}\,. \tag{1.34}$$

1.4 Anti-de Sitter space

Anti-de Sitter space in $(d+1)$-dimensional spacetime dimensions, which we denote AdS (or AdS_{d+1} if we want to emphasise the number of dimensions), has a metric which may be written as

$$\mathrm{d}s^2 = \frac{r^2}{L^2}\eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu + \frac{L^2}{r^2}\,\mathrm{d}r^2\,, \tag{1.35}$$

where μ,ν run from 0 to $d-1$, and $\eta_{\mu\nu}$ is the d-dimensional Minkowski metric. If you compute the Ricci tensor of AdS you will find that it is proportional to the metric and that the Ricci scalar is constant,

$$R_{\mu\nu} = -\frac{d}{L^2}g_{\mu\nu}\,, \qquad R = -\frac{d(d+1)}{L^2}\,. \tag{1.36}$$

We therefore find that AdS satisfies the vacuum Einstein equations with a negative cosmological constant Λ related to L,

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + g_{\mu\nu}\Lambda = 0\,, \qquad \Lambda = -\frac{d(d-1)}{2L^2}\,. \tag{1.37}$$

The coordinate r runs over all positive real values. At $r \to \infty$ one reaches the conformal boundary of AdS. It is not a real boundary, since it is at infinite distance from any point at finite r. Despite this, it is common to drop the word "conformal" and simply refer to $r \to \infty$ as *the boundary* of AdS. The coordinate singularity at $r \to 0$ is known as the *Poincaré horizon.*

Anti-de Sitter space is an example of a maximally symmetric spacetime, meaning that it has the maximum possible number of independent Killing vectors. In $(d+1)$-dimensions this corresponds $(d+1)(d+2)/2$ Killing vector fields. To express the isometries generated by these Killing vectors, it will be convenient to perform a coordinate transformation, replacing r with $z = L^2/r$. In terms of z, the metric of AdS is

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left(\eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu + \mathrm{d}z^2\right), \tag{1.38}$$

In these coordinates the boundary is at $z = 0$, while the Poincaré horizon is at $z \to \infty$.

What are the isometries of the metric in equation (1.38)? Isometries we can write down immediately are Lorentz transformations and rotations in the x^μ coordinates,

$$x^\mu \to R^\mu{}_\nu x^\nu + a^\mu \qquad z \to z\,, \tag{1.39}$$

where $R^\mu{}_\nu$ is a constant $\mathrm{SO}(d-1,1)$ matrix and a^μ is a constant d-dimensional vector. Since translations and rotations leave the Minkowski metric invariant and we are not touching z, these transformations leave the metric in equation (1.38) invariant. Another isometry of AdS is the scale transformation

$$x^\mu \to \alpha x^\mu\,, \qquad z \to \alpha z\,, \tag{1.40}$$

for arbitrary postive constant α. To write the final set of isometries, let us define $X^m = (x^\mu, z)$ and a constant vector $b^m = (b^\mu, 0)$. The isometry is then

$$X^m \to \frac{x^m + b^m(X\cdot X)}{1 + 2\,b\cdot X + (b\cdot b)(X\cdot X)}\,, \tag{1.41}$$

where the dot denotes contraction with the $(d+1)$-dimensional Minkowski metric.

Together, all of these isometries form the group $\mathrm{SO}(d,2)$. This is the same as the conformal group in d dimensions. Indeed, notice that the transformation of the x^μ coordinates in equations (1.39) and (1.40), and the $z \to 0$ limit of (1.41), together match the conformal transformations described in the previous section.

Another common coordinate choice for AdS are *global coordinates*, in which AdS_{d+1} is

$$\mathrm{d}s^2 = -\left(1 + \frac{\rho^2}{L^2}\right)\mathrm{d}t^2 + \frac{\mathrm{d}\rho^2}{1+\rho^2/L^2} + \rho^2\,\mathrm{d}\Omega_{d-1}^2\,, \tag{1.42}$$

where we use $\mathrm{d}\Omega_{d-1}^2$ to denote the metric on a unit, round $(d-1)$-dimensional sphere. Introducing angular coordinates θ_1, θ_2, up to θ_{d-1}, this metric may be written

$$\mathrm{d}\Omega_{d-1}^2 = \mathrm{d}\theta_1^2 + \sin^2\theta_1\,\mathrm{d}\theta_2^2 + \sin^2\theta_1\sin^2\theta_2\,\mathrm{d}\theta_3^2 + \cdots\,. \tag{1.43}$$

All of the angles take values in the range $\theta_i \in [0,\pi]$, except for $\theta_{d-1} \in [0, 2\pi]$. The space parameterised by global coordinates is often referred to as *global AdS*, or simply *AdS*. The coordinates used in the metric (1.35) or (1.38) actually cover only part of global AdS. For this reason, the space covered by the coordinates in equation (1.35) is more precisely referred to as *the Poincaré patch* of AdS.

1.5 Black hole thermodynamics

Black holes are thermodynamic objects. For example, by considering quantum fields on a black hole spacetime one can show that they emit radiation with a blackbody spectrum. The temperature of this blackbody spectrum is known as the Hawking temperature. We may calculate the Hawking temperature by a purely classical calculation, which we will illustrate for the example of a Schwarzschild black hole.

The metric of a Schwarzschild black hole with Schwarzschild radius r_s is

$$\mathrm{d}s^2 = -f(r)\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2)\,, \qquad f(r) = 1 - \frac{r_s}{r}\,. \tag{1.44}$$

If we Wick rotate to imaginary time τ, this metric becomes

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}\tau^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2)\,. \tag{1.45}$$

Recall that at finite temperature the imaginary time is periodic with period equal to the inverse temperature, which we denote $\tau \sim \tau + \beta$. We will show that there is a specific value of β that avoids a singularity in the metric at $r = r_s$.

If we expand the metric in equation (1.45) near $r = r_s$, we find

$$\mathrm{d}s^2 \approx f'(r_s)(r - r_s)\,\mathrm{d}\tau^2 + \frac{\mathrm{d}r^2}{f'(r_s)(r - r_s)} + \cdots\,, \tag{1.46}$$

where the dots denote higher powers in the $r - r_s$ expansion, plus the $\mathrm{d}\theta^2$ and $\mathrm{d}\phi^2$ terms in the metric which will play no role in this calculation. For the next step, define a new coordinate ρ that satisfies

$$\mathrm{d}\rho = \frac{\mathrm{d}r}{\sqrt{f'(r_s)(r - r_s)}} \qquad \Rightarrow \qquad \rho = \frac{2\sqrt{r - r_s}}{\sqrt{f'(r_s)}}\,. \tag{1.47}$$

In terms of ρ, the metric in equation (1.46) becomes

$$\mathrm{d}s^2 = \mathrm{d}\rho^2 + \frac{(f'(r_s))^2}{4}\rho^2\,\mathrm{d}\tau^2 + \cdots\,. \tag{1.48}$$

Finally, define a rescaled Euclidean time coordinate $\alpha = f'(r_s)\tau/2$, turning the metric into

$$\mathrm{d}s^2 = \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\alpha^2 + \cdots\,. \tag{1.49}$$

This looks like the metric of a plane, written in polar coordinates. However, it only describes a plane if $\alpha \sim \alpha + 2\pi$. If the period of α is different from 2π the metric instead describes a cone, with a conical singularity at the tip of the cone at $\rho = 0$. Thus, to avoid the conical singularity we demand $\alpha \sim \alpha + 2\pi$, and therefore $\tau \sim \tau + 4\pi/f'(r_s)$. Identifying the period of τ with the inverse temperature $\beta = 1/T$, we can read off the Hawking temperature

$$T = \frac{f'(r_s)}{4\pi} = \frac{1}{4\pi r_s}\,. \tag{1.50}$$

There is a whole theory of black hole thermodynamics, where for example the energy is identified with the mass of the black hole. Later we will see how this works in asymptotically-AdS spaces. A crucial result that we will need is that a black hole with a horizon of surface area A has entropy S, known as the Bekenstein–Hawking entropy, given by the formula

$$S = \frac{A}{4G}\,, \tag{1.51}$$

where G is Newton's constant. For example, the Schwarzschild black hole in equation (1.44) has $A = 4\pi r_s^2$, and hence $S = \pi r_s^2/G$. Recalling that the Schwarzschild radius is related to the mass M of the black hole by $r_s = 2GM$ and identifying the mass as the energy E, we ind

$$E = \frac{1}{2}\sqrt{\frac{S}{\pi G}} \qquad \Rightarrow \qquad \frac{\partial E}{\partial S} = \frac{1}{4\sqrt{\pi G S}} = \frac{1}{4\pi r_s} = T\,. \tag{1.52}$$

We see that the Hawking temperature is the derivative of the energy with respect to entropy, in accordance with usual thermodynamics.

1.6 Why might gravity be holographic?

Remarkably, thought experiments suggest that the Bekenstein–Hawking entropy should provide an upper bound on the entropy of a region of spacetime in any quantum theory of gravity. For example, consider a spherical region Γ of spacetime, with surface area A and containing a thermodynamic system with some initial entropy S_i. The mass of this thermodynamic system must not be too large, since if we try to fit a very large mass into our region then it would form a black hole larger than Γ.

Now imagine we drop a spherical shell of matter into Γ, choosing the mass of the shell to be such that when added to the mass of the thermodynamic system we end up with a black hole that exactly fills Γ. At the end of this process, the final entropy S_f is then the entropy of a black hole of area A, given by equation (1.51). But the second law of thermodynamics tells us that entropy before should be smaller than the entropy after, $S_i \leq S_f$, and hence

$$S_i \leq \frac{A}{4G}\,. \tag{1.53}$$

Thus, the Bekenstein–Hawking entropy provides an upper bound on the amount of entropy that can be present in Γ.

What is remarkable about the bound in equation (1.53) is that entropy is usually an extensive quantity, i.e. one that scales as the volume of the system. For instance, consider spin-half particles located at the vertices a simple cubic lattice, with lattice spacing a and each side of length $V^{1/3} \gg a$, as depicted in figure 1. The volume of the lattice is then V, and the total number of particles is $N = V/a^3$. There are two independent states associated to each lattice site, spin up and spin down. The dimension of the Hilbert space, i.e. the total number of independent states of on the lattice, is then $\dim \mathcal{H} = 2^N$. The entropy of a macrostate is determined by the number Ω of microstates contributing to that macrostate by $S = \log \Omega$. Since the number of microstates cannot be larger than the total number of states, $\Omega \leq N = V/a^3$, we find that the entropy of any macrostate is bounded from above,

$$S \leq \frac{V}{a^3}\log 2\,. \tag{1.54}$$

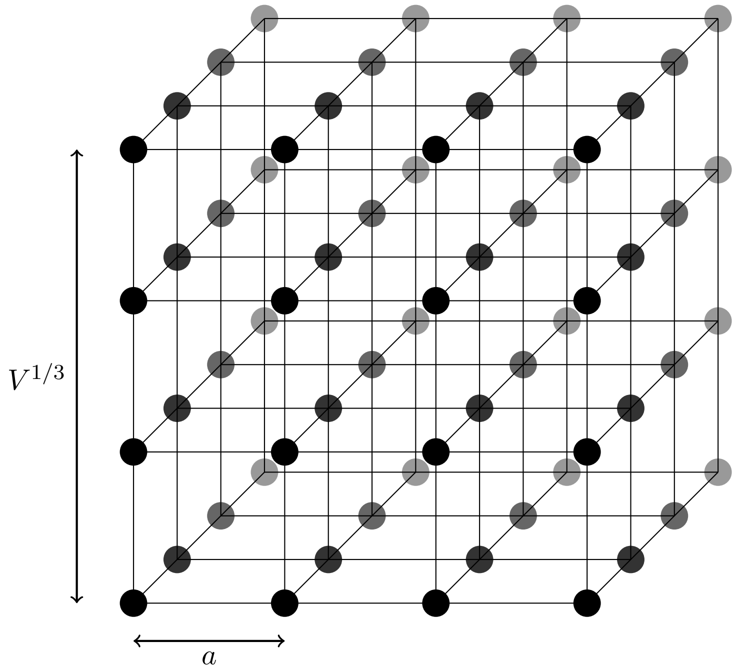


Figure 1: A simple cubic lattice of spins.

The upper bound on entropy for gravity in equation (1.53), set by area rather than volume, provides a hint that the degrees of freedom in quantum gravity should somehow live in one lower dimension. This is known as the holographic principle, after the way a hologram encodes a 3D image on a 2D surface. The AdS/CFT correspondence, which is the topic of these lectures, is an example of a concrete realisation of this idea.

2 Elements of string theory

Holography emerged as a conjecture from string theory. In this section we will state some basic facts about string theory needed to motivate the conjecture.

2.1 Bosonic string theory

As the name implies, the fundamental objects of string theory are *strings*, $(1+1)$-dimensional objects. The simplest form of string theory is bosonic string theory, for which the action of a string is proportional to the area of its worldsheet. This action is called the Nambu–Goto action. Suppose a string propagates in a D-dimensional Minkowski spacetime with coordinates x^M and metric g_{MN}. Paramaterising the worldsheet by coordinates σ^0 and σ^1, the embedding of the string in D-dimensional spacetime is specified by $x^M = X^M(\sigma)$ for some functions X^M. The Nambu–Goto action is then, in Euclidean signature

$$I_{\mathrm{NG}} = \frac{1}{2\pi\alpha'} \int \mathrm{d}^2\sigma \sqrt{\det(g_{MN}\partial_a X^M \partial_b X^N)}\,, \tag{2.1}$$

where $\partial_a \equiv \partial/\partial\sigma^a$, and α' is a parameter of dimension [length]^2, called the Regge slope. The typical size of strings is $\ell_s = \sqrt{\alpha'}$, called the string length.

Classically, a string can vibrate. Quantum mechanically, these vibrations are quantized. The energies of these vibrational quanta are the masses of the corresonding states. It turns out that quantization of the bosonic string is only consistent with Lorent-invariance of D-dimensional Minkowski space if $D = 26$. The square root in the Nambu–Goto action makes it difficult to quantize, so it is typical to instead work with the Polyakov action,[2] in which one introduces an auxiliary field γ_{ab} called the world-sheet metric

$$I_{\mathrm{P}} = \frac{1}{4\pi\alpha'} \int \mathrm{d}^2\sigma \sqrt{\gamma}\, \gamma^{ab} g_{MN} \partial_a X^M \partial_b X^N \,, \tag{2.2}$$

where γ^{ab} is the inverse of γ_{ab} and $\gamma \equiv \det \gamma_{ab}$. If you solve Euler–Lagrange equation for γ_{ab} and substitute the solution into the Polyakov action then you recover the Nambu–Goto action, so the two actions are equivalent at the classical level.

Strings can be either open or closed. We consider first closed strings; we'll come to open strings later. In the quantization of the bosonic closed string the lowest energy state turns out to be a Lorentz scalar with mass-squared given by

$$m^2 = -\frac{1}{\alpha'} \,. \tag{2.3}$$

This negative m^2 means the particle is a tachyon, and signals an instability of the theory. This is bad, but for now let's ignore the problem and look at the first excited level, where it turns out that one finds massless states. Under Lorentz transformations one of these states transforms as Lorentz scalar, some of them transform as an antisymmetric tensor, and the rest as a symmetric tensor. The latter is what makes string theory a theory of quantum gravity; the massless symmetric tensor corresponds to gravitons, linearised fluctuations of the metric.

Adding further excitations, one obtains an infinite tower of states of various spins, with positive mass-squared $m^2 = n/\alpha'$ for positive integer n.

String interactions. There is a term that can be added to the Polyakov action that is consistent with all of its symmetries, consisting of the integral of the Ricci scalar R of the string world sheet with some coefficient. With this term, the action for the string in flat space becomes

$$I = I_{\mathrm{P}} + \frac{c}{4\pi} \int \mathrm{d}^2\sigma \sqrt{\gamma}\, R \,, \tag{2.4}$$

for some constant c. Adding this term does not affect the equations of motion because in two-dimensions the integral of the Ricci scalar is topological, it computes the Euler characteristic of the string worldsheet,

$$\frac{1}{4\pi} \int \mathrm{d}^2\sigma \sqrt{\gamma}\, R = \chi = 2 - 2\mathrm{g} \,, \tag{2.5}$$

where on the right we have used the relation between the Euler characteristic and the genus g. As a consequence of this term, string theory has a perturbative expansion in increasing worldsheet genus. For example, in figure 2 we sketch the diagrammatic expansion for the vacuum amplitude. Similar to Feynman diagrams in QFT, there is a procedure to

[2] Introduced by Brink, Di Vecchia, Howe, Deser, and Zumino. It was used by Polyakov to quantize the string.

associate an integral to each of the diagrams in the figure. In particular a diagram of genus g is proportional to $e^{2c(\mathrm{g}-1)}$, with exponential appearing because the Euclidean action is exponentiated in the pather integral. If we define the *string coupling* $g_s = \log c$, this becomes $g_s^{2\mathrm{g}-2}$. We see that the perturbative expansion makes sense if $g_s \ll 1$.

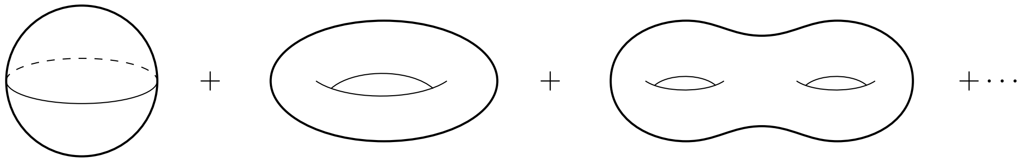


Figure 2: The perturbative expansion in string theory is an expansion in increasingly higher genus Riemann surfaces. The figure shows the closed string vacuum diagrams.

We see here a hint that string theory could be related to large-N gauge theory. Recall that we could associate a genus to the Feynman diagrams in the double-line notation in Yang–Mills theory, and that at large N a diagram scales as $N^{2-2\mathrm{g}}$. This suggests an analogy between the perturbative expansion of string theory and the large-N expansion of gauge theory, with $g_s \propto N^{-1}$. AdS/CFT will make this precise.

The low energy effective theory. Suppose we are only interested in physics at length scales much larger than the string length, $\ell \gg \sqrt{\alpha'}$, or equivalently at energy scales much lower than the mass of the massive string states, $E \ll 1/\sqrt{\alpha'}$. Then only the massless states are relevant. Each of these massless states give rise to a field. The scalar states give rise to a scalar field called the dilaton ϕ. The antisymmetric tensor states give rise to an antisymmetric tensor field $B_{\mu\nu}$ and, as alluded to earlier, the symmetric tensor states give rise to the metric $g_{\mu\nu}$.

There is a procedure for working out from string theory the equations of motion for the fields describing the massless string states. One can then work out the action that gives rise to these equations of motion. It turns out to be

$$S = \frac{1}{16\pi G}\int \mathrm{d}^{26}x\,\sqrt{-g}\,e^{-2\phi}\left(R - \frac{1}{12}H_{\mu\nu\lambda}H^{\mu\nu\lambda} + 4\partial_\mu\phi\partial^\mu\phi\right). \tag{2.6}$$

The strength of interactions is set by Newton's constant G, which is related to the string coupling by $G \propto g_s^2$.

Open strings and D-branes. Now let us briefly consider open strings. Since an open string has endpoints, we need to impose boundary conditions on the fields X^M. We have two choices; Neumann or Dirichlet. If we choose a Neumann boundary condition for a given M then the endpoint of the string is free to move about in this direction. On the other hand, if we impose a Dirichlet boundary condition then the endpoint of the string is fixed to a particular point in this direction.

In total, if we choose some directions to have Neumann boundary conditions and the rest to have Dirichlet, then the endpoint of the string will be fixed to a surface, which is called a *D-brane*. See figure 3. Often, we want to emphasise the dimensionality of the D-brane (i.e. the number of Neumann directions), so a $(p+1)$-spacetime dimensional D-brane is called a Dp-brane.

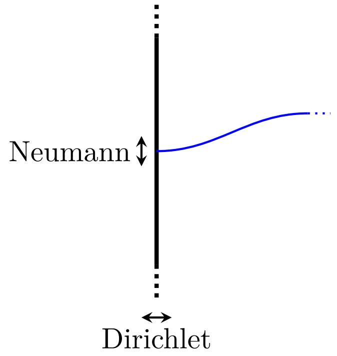


Figure 3: The vertical black line represents a D-brane. The blue curve is an open string with one endpoint on the D-brane. The endpoint is fixed to the D-brane, so it satisfies Dirichlet boundary conditions in the directions perpendicular to the D-brane and Neumann boundary conditions in the parallel directions.

Quantizing the open string in the presence of a D-brane reveals that D-branes are themselves dynamical objects. In particular, an open string in the presence of a Dp-brane has massless states corresponding to a $(p+1)$-dimensional vector field. The D-brane is a $(p+1)$-dimensional gauge theory with gauge group U(1). If we bring N Dp-branes together to form a "stack", i.e. so that they sit on top of each other the gauge group is instead U(N).

2.2 Superstring theory

The tachyon present in bosonic string theory can be eliminated by modifying string theory to be supersymmetric. This can be done by adding some fermionic degrees of freedom to the Polyakov action. One effect of this is to lower the number of spacetime dimensions in which string theory is consistent with Lorentz invariance from $D = 26$ to $D = 10$. There are several different ways of making strings supersymmetric, leading to ten-dimensional string theories with different names. For these lectures we'll focus on one, called *type IIB string theory*.

When the quantization of type IIB string theory is carried out, one finds that there is no tachyon. There are still the same massless closed string states as before, giving rise to a dilaton ϕ, an antisymmetric tensor B_{MN}, and the metric g_{MN}. Moreover, there are additional massless states, corresponding to another scalar $C_{(0)}$, a two-form gauge field (i.e. an antisymmetric rank-two tensor) $C_{(2)}$, and a four-form gauge field $C_{(4)}$. There are also massless fermionic states, since supersymmetry requires that for every bosonic degree of freedom there must be a fermionic one, and vice-versa. The low-energy effective theory describing these fields is a supergravity theory called type IIB supergravity.

In type IIB string theory there can only be Dp-branes with even p. Just like in bosonic string theory, the low energy theory describing a stack of D-branes that arises from quantizing the open string is a Yang–Mills theory, this time a supersymmetric one. Most importantly for these lectures:

The low energy theory describing a stack of D3-branes is $\mathcal{N}=4$ supersymmetric Yang–Mills theory. The coupling constant g_{YM} is related to the string coupling by

$$g_{\mathrm{YM}}^2 = 4\pi g_s\,. \tag{2.7}$$

An important property of Dp-branes is that they are charged under the gauge fields $C_{(p+1)}$, so for example a D3-brane is charged under the gauge field $C_{(4)}$.

D-branes also arise in supergravity, as solitonic solutions to the equations of motion. For example, the D3-brane solution of type IIB supergravity has a metric of the form

$$\mathrm{d}s^2 = H(r)^{-1/2}\eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu + H(r)^{1/2}\left(\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega_5^2\right)\,, \tag{2.8}$$

where $\eta_{\mu\nu}$ is the four-dimensional Minkowski metric, $\mathrm{d}\Omega_5^2$ is the metric ona unit S^5, and

$$H(r) = 1 + \frac{L^4}{r^4}\,, \tag{2.9}$$

for a length scale L. The other type IIB supergravity fields vanish in this solution, except for the dilaton, which is constant, and the four-form gauge field $C_{(4)}$, the form of which may be found for example in ref. [1]. This four-form field is sourced by the D3-branes, and one can compute the number N of D3-branes by integrating the corresponding five-form field strength over the five-sphere. The result is

$$N = \frac{L^4}{4\pi g_s \alpha'^2} \tag{2.10}$$

For the supergravity description to be reliable we need L, the typical length scale of this solution, to be much larger than $\sqrt{\alpha'}$, the typical length scale of strings. From equation (2.10) we see that $L \gg \sqrt{\alpha'}$ translates to $g_s N \gg 1$.

Closed strings propagating in the *near-horizon* part of the spacetime, corresponding to $r \ll L$, are redshifted. In the near-horizon limit we have that $H(r) \approx L^4/r^4$, and thus the metric in equation (2.8) becomes in the near-horizon limit

$$\mathrm{d}s^2 \approx \frac{r^2}{L^2}\eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu + \frac{L^2}{r^2}\,\mathrm{d}r^2 + L^2\,\mathrm{d}\Omega_5^2\,. \tag{2.11}$$

This is precisely the metric of $\mathrm{AdS}_5 \times \mathrm{S}^5$, where both AdS_5 and S^5 have the same curvature radius L. Consequently, the low-energy physics of the D3-branes can be described by type IIB supergravity in this $\mathrm{AdS}_5 \times \mathrm{S}^5$ geometry.

3 The AdS/CFT correspondence

3.1 The conjecture

In the previous section we saw two descriptions of the low energy physics of a stack of N D3-branes. Both rely on string theory to be perturbative, $g_s \ll 1$, in order to be reliable. The two descriptions are:

- The open string description. In this description the D3-branes are described by $\mathcal{N} = 4$ SYM with gauge group $\mathrm{SU}(N)$ and gauge coupling $g^2_{\mathrm{YM}} = 4\pi g_s$,[3] or equivalently with 't Hooft coupling
$$\lambda = 4\pi g_s N\,. \tag{3.1}$$
This decription is known to be reliable if the gauge theory is perturbative, $\lambda \ll 1$ or equivalently $g_s N \ll 1$.
- The closed string description, in which the D3-branes are described by type IIB supergravity on $\mathrm{AdS}_5 \times \mathrm{S}^5$, where both $\mathrm{AdS}_5 \times \mathrm{S}^5$ have radius L given by
$$L^4 = 4\pi g_s N \alpha'^2\,. \tag{3.2}$$
As discussed earlier, this perspective is reliable as long as the curvature of the spacetime is small compared to the string scale, $L^2 \gg \alpha'$, so that stringy corrections to supergravity are small, corresponding to $g_s N \gg 1$.

We see that the two descriptions are known to be reliable in opposite regimes of $g_s N$. The AdS/CFT conjecture is that the descriptions are valid beyond these regimes. Concretely, suppose we start at $g_s N \ll 1$, where the D3-branes can be described by $\mathcal{N} = 4$ SYM with 't Hooft coupling $\lambda = 4\pi g_s N \ll 1$. Now imagine dialing up λ. Yang–Mills theory makes sense at all values of the 't Hooft coupling, so it's not obvious what should go wrong with this description of D3-brane physics in the limit where $\lambda = 4\pi g_s N \gg 1$. This physical argument leads to the conjecture that:

AdS/CFT conjecture: Four-dimensional $\mathcal{N} = 4$ supersymmetric Yang–Mills theory with gauge group $\mathrm{SU}(N)$, in the limit $N \to \infty$ and at strong 't Hooft coupling $\lambda \gg 1$, is equivalent to type IIB supergravity on $\mathrm{AdS}_5 \times \mathrm{S}^5$. We say that the two theories are *holographically dual* to one another.

The equivalence can be stated as the fact that the generating functionals of the two theories are equal, and leads to "holographic dictionary" — a map between observables in the two theories. In these lectures we will get into some of the nuts and bolts of what the equivalence means and how it can be useful. Part of the power of AdS/CFT is that the conjecture applies when $\lambda \gg 1$, i.e. when $\mathcal{N} = 4$ SYM strongly coupled. This means that we can use it to study a strongly coupled QFT via calculations in classical gravity!

As a first consistency check, if two theories are equivalent then their global symmetries should be the same. The bosonic global symmetries on the gravitational side are the isometries of the $\mathrm{AdS}_5 \times \mathrm{S}^5$ spacetime, namely $\mathrm{SO}(4,2) \times \mathrm{SO}(6)$. Being a four-dimensional CFT, $\mathcal{N} = 4$ SYM is invariant under the four-dimensional conformal group $\mathrm{SO}(4,2)$ and as we previously mentioned there is an also an $\mathrm{SO}(6)$ symmetry that mixes the six scalars, so the global symmetries of $\mathcal{N} = 4$ are also $\mathrm{SO}(4,2) \times \mathrm{SO}(6)$. So the global symmetries do indeed match.

Although there is no proof that the AdS/CFT conjecture is true, there is by now a large body of evidence that it is indeed correct. Tests of the conjecture typically work by computing a physical quantity both using $\mathcal{N} = 4$ SYM and type IIB supergravity

[3] Actually, we previously stated that the gauge group was $\mathrm{U}(N)$. There is a $\mathrm{U}(1)$ factor that decouples from the $\mathrm{SU}(N)$ fields and is usually dropped. You can read more about this in section 5.2.3 of ref. [1].

and checking that the two results agree. This can be a challenge since when $\lambda \gg 1$ we cannot typically use perturbation theory on the QFT side. However, the large amount of symmetry of $\mathcal{N} = 4$ SYM (maximal supersymmetry plus conformal symmetry) helps with the tests, as for example it implies that some observables are one-loop exact. Such observables can be computed using perturbation theory at weak coupling with the resulting formulas also hold at strong coupling. Moreover there are some observables that can be computed nonperturbatively. So far, AdS/CFT has passed every test.

It's worth mentioning that the form of the AdS/CFT conjecture we have stated is the weakest form of the AdS/CFT conjecture. There are also stronger forms of the conjecture which relax the requirements $\lambda \gg 1$ and possibly also $N \to \infty$ in $\mathcal{N} = 4$ SYM, replacing the supergravity with string theory. We will focus on the weak form in the box, since it is the best evidenced and easiest to understand.

As we will see when we explore how the correspondence works, it is natural to the think of the $\mathcal{N} = 4$ SYM as "living" at the conformal boundary of AdS_5. In this way, AdS/CFT provides a concrete realisation of the holographic principle: we can describe what is happening in the bulk of AdSusing degrees of freedom localised to the boundary.

When comparing $\mathcal{N} = 4$ SYM and supergravity/string theory on $\mathrm{AdS}_5 \times \mathrm{S}^5$, we will need to use relations between the parameters in the two theories. Comparing equations (3.1) and (3.2), we see that the 't Hooft coupling is related to the curvature radius of $\mathrm{AdS}_5 \times \mathrm{S}^5$ and the Regge parameter by

$$L^4 = \lambda \alpha'^2 \,. \tag{3.3}$$

The supergravity theory has a Newton constant, which we denote $G_{(10)}$ where the 10 stands for the number of dimensions. It is related to the Regge slope and the string coupling by $G_{(10)} = 8\pi^6 \alpha'^4 g_s^2$. If we use equation (3.2) to express this in terms of L instead of N, we find

$$G_{(10)} = \frac{\pi^4 L^8}{2N^2} \,. \tag{3.4}$$

3.2 Other examples of holography

The equivalence between $\mathcal{N} = 4$ SYM and type IIB supergravity is the most well-studied and understood example of a pair of theories related by AdS/CFT, but far from the only one. There are many other examples of similar dualities between d-dimensional CFTs and a gravitational theories on AdS_{d+1}, or on $\mathrm{AdS}_{d+1} \times \mathcal{M}$ where $\mathcal{M}$ is some compact manifold. There are also examples where the dual theory is not conformal, in which case the AdS_{d+1} in the dual gravitational theory is replaced with a spacetime which is only asymptotically AdS. Most examples are motivated by physical arguments from string theory. General principles are that the classical limit on the gravity side corresponds to a limit of many degrees of freedom (like large N in SYM) on the QFT side, and ignoring stringy corrections corresponds to a strong coupling limit in the QFT.

The need for a string theory description is quite constraining, in the sense that if you are handed some QFT then it's very hard to say what the gravitational dual is, assuming a gravitational dual even exists. Similarly, given a theory of gravity in (asymptotically) AdS spacetime, it is hard to say what the dual QFT is. This is a problem if you want to use holography to model a particular system described strongly coupled QFT. A common approach in research is *bottom-up* holography, in which you build a toy model of strongly coupled QFT by writing down a gravitational action without knowing the dual QFT, tuning the gravitational theory to reproduce features of interest in the QFT.

In the next subsection we will be agnostic about the particular QFT and gravitational dual with which we work. We will then return to $\mathcal{N}=4$ SYM and type IIB supergravity in subsequent examples.

3.3 The field/operator map

For any QFT and its gravitational dual, AdS/CFT is an equality of generating functionals. Let us denote the generating functional of the QFT as $Z_{\mathrm{QFT}}[J]$. It is a functional of sources J. We denote the generating functional for the dual gravitational theory as $Z_{\mathrm{grav}}[J]$. We will always work in the classical limit on the gravitational side, so $Z_{\mathrm{grav}}[J] \approx e^{iS^\star_{\mathrm{grav}}[J]}$, where $S^\star_{\mathrm{grav}}[J]$ is the on-shell action of the gravitational theory. Thus, AdS/CFT is the statement:

$$Z_{\mathrm{QFT}}[J] = e^{iS^\star_{\mathrm{grav}}[J]} \tag{3.5}$$

But what is the meaning of J in the gravity side? The answer turns out to be boundary conditions. For every gauge-invariant operator $\mathcal{O}$ in the QFT, there is a field ϕ of the same spin in the dual gravitational theory. This field will obey some classical equations of motion, and requires a boundary condition at the conformal boundary of AdS. Roughly speaking $J = \phi|_{\mathrm{boundary}}$ (we will make this more precise soon).

Equation (3.5) allows us to use the gravitational theory to compute correlation functions in the dual QFT. The general procedure is that one should solve the equations of motion of the gravitational theory with the appropriate boundary conditions, substitute the solution into the action, and then differentiate to obtain the correlation functions. For example, equation (3.5) implies that the generating functional of connected correlation functions in the QFT is $W[J] = S^\star_{\mathrm{grav}}$, so the connected correlation functions may be computed by the functional derivatives

$$\begin{aligned} \langle \mathcal{O}(x) \rangle_J &= \frac{\delta S^\star_{\mathrm{grav}}[J]}{\delta J(x)}\,, \\ \langle \mathcal{O}(x)\mathcal{O}(y) \rangle_{\mathrm{c},J} &= \frac{\delta^2 S^\star_{\mathrm{grav}}[J]}{\delta J(x)\,\delta J(y)}\,, \end{aligned} \tag{3.6}$$

and so-on.

3.4 Scalars

Let's take a look at the field/operator map in more detail. Suppose we have a $(d+1)$-dimensional gravitational theory consisting of just the metric g_{mn} and a scalar field ϕ with mass m, with a negative cosmological constant. We take the action to be

$$\begin{aligned} S_{\mathrm{grav}} &= \frac{1}{16\pi G}\int \mathrm{d}^{d+1}x\sqrt{-g}\,\left(R + \frac{d(d-1)}{L^2}\right) + S_\phi\,, \\ S_\phi &= -\frac{1}{2}\int \mathrm{d}^{d+1}x\sqrt{-g}\,\left((\partial\phi)^2 + m^2\phi^2\right)\,. \end{aligned} \tag{3.7}$$

This is example where we don't know what the dual QFT is. However, for various choices of d and m this action will appear within a larger action of a theory for which we *do* know the dual QFT, so the analysis we perform here will have some general applicability.

The Einstein equations following from the action in equation (3.7) admit a solution where $\phi = 0$ and the metric is that of AdS_{d+1}. We will work in Poincaré coordinates,

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left(\mathrm{d}z^2 + \eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu\right). \tag{3.8}$$

Now let's turn on the scalar field. For simplicity, we will work in a *probe limit* where the value of the scalar field we turn on is small enough that it doesn't backreact significantly on the metric. In other words, we will keep the metric fixed as AdS_{d+1} in equation (3.8) and solve only for the scalar field.

The equation of motion for the scalar field following from the action in equation (3.7) is

$$\frac{1}{\sqrt{-g}}\partial_m\left(\sqrt{-g}\,g^{mn}\partial_n\phi\right) - m^2\phi = 0\,. \tag{3.9}$$

Plugging in the metric from equation (3.8) we find that this equation of motion becomes

$$z^2\phi'' - (d-1)z\phi' + z^2\eta^{\mu\nu}\partial_\mu\partial_\nu\phi - m^2L^2\phi = 0\,, \tag{3.10}$$

where the primes denote derivatives with respect to z. We need to know what kind of boundary conditions we can impose at $z = 0$. To this end, we use the Frobenius method. The first step is to look for an approximate solution near $z = 0$ by substituting the ansatz $\phi(z,x) = c(x)z^\gamma$ into the equation of motion, for some function $c(x)$ of the field theory directions and some number γ. One finds that the equation of motion becomes

$$\left[\gamma(d-\gamma) - m^2L^2\right]c(x) + z^2\eta^{\mu\nu}\partial_\mu\partial_\nu c(x) = 0\,. \tag{3.11}$$

At leading order in small z, we can drop the term involving derivatives of $c(x)$, and we learn that the allowed values of γ satisfy $\gamma(\gamma - d) = m^2L^2$. The two solutions to this quadratic equation correspond to the two independent solutions of the second-order ODE. Let us denote the largest solution of this equation as $\gamma = \Delta$. The other solution is then $\gamma = d - \Delta$. The Frobenius method then tells us that the general solution to the equation of motion (3.10) has a near-boundary expansion of the form

$$\phi(z,x) = z^{d-\Delta}\left[c_1(x) + \cdots\right] + z^\Delta\left[c_2(x) + \cdots\right]\,, \tag{3.12}$$

with functions $c_1(x)$ and $c_2(x)$ to be determined by the boundary conditions, and where the dots denote terms containing higher powers of z. The coefficients of the higher powers of z are determined in terms of $c_1(x)$ and $c_2(x)$ by the equations of motion. Important AdS jargon is that the solution proportional to z^Δ is known as the *normalisable* solution, while the solution proportional to $z^{d-\Delta}$ is known as the *non-normalisable solution.*

Recall that AdS has a scaling isometry, under which $z \to \alpha z$ and $x^\mu \to \alpha x^\mu$ for any constant α, which from the boundary looks like a scale transformation. The solution for $\phi(x,z)$ in equation (3.12) is invariant under this transformation if we also rescale the boundary conditions, as $c_1 \to \alpha^{-(d-\Delta)}c_1$ and $c_2 \to \alpha^{-\Delta}c_2$. The transformation of c_2 is precisely the scaling behaviour of a scalar operator $\mathcal{O}$ of conformal dimension Δ. In order to make the integral $\int \mathrm{d}^d x\, J(x)\mathcal{O}(x)$ dimensionless, and operator of dimension Δ should have a source of dimension $d - \Delta$, which precisely matches the scaling behaviour of c_1. This leads us to the precise statement of the identification between boundary conditions and sources; we should identify $J(x) = c_1(x)$. This deserves a box:

A scalar field ϕ with mass m in AdS_{d+1} with curvature radius L is dual to an operator of dimension Δ, where

$$\Delta(\Delta - d) = m^2 L^2 \,. \tag{3.13}$$

We identify the the source $J(x)$ for the scalar operator with the coefficient of the non-normalisable solution for the scalar field, which can be extracted from the limit

$$J(x) = \lim_{z\to 0} z^{\Delta - d}\phi(z, x) \,. \tag{3.14}$$

These relations also hold in asymptotically AdS spacetimes. Notice from equation (3.13) that a relevant operator, which has $\Delta < d$, is dual to a scalar field with negative m^2. This is okay because in AdS such a scalar field is stable as long as its mass obeys the *Breitenlohner–Freedman* bound

$$m^2 L^2 \geq -\frac{d^2}{4} \,. \tag{3.15}$$

The fact that the coefficient of the normalisable solution c_2 scales in the same way as $\mathcal{O}$ is suggestive that $c_2 \propto \langle \mathcal{O} \rangle$. Indeed, this typically turns out to be the case when you calculate $\langle \mathcal{O} \rangle$ using the prescription in equation (3.6).

If we have a solution for $\phi(z, x)$ that behaves near the boundary as $\phi(z, x) \approx J(x) z^{d-\Delta}$, the next step to obtain correlation functions should be to substitute the solution into the action to obtain the on-shell action. However, if we evaluate the Lagrangian density of the scalar field part of the action (3.7) near $z = 0$, we find

$$-\frac{1}{2}\left((\partial\phi)^2 + m^2\phi^2\right) \approx -\frac{1}{2} J(x)^2 L^{d-1} z^{d-1-2\Delta}(d - 2\Delta)(d - \Delta) + \cdots \,, \tag{3.16}$$

where the dots denote terms of higher order in a small-z expansion. We see that the Lagrangian density blows up near $z = 0$, so if we integrate over all of z the action will diverge. To obtain a finite result for the on-shell action that we can use in the formulas (3.6) to calculate correlation functions, we regularise the action by integrating only over $z \geq \epsilon$ for some small length ϵ. The action will be finite for finite ϵ, and diverge if we send $\epsilon \to 0$. We then eliminate the divergence by adding *counterterms* to the action. These are boundary terms evaluated on the cutoff surface at $z = \epsilon$. Since they are boundary terms, they do not change the equations of motion. Their only role is to render the on-shell action finite.

The form of the counterterms depends on d and Δ, so we will not be able to write a general formula for them. However to illustrate the procedure we will determine down one counterterm which is generically present for scalar fields. First notice that we can integrate by parts in the scalar field action (3.7),

$$S_\phi = -\frac{1}{2}\int \mathrm{d}^{d+1}x\sqrt{-g}\left(\nabla^m(\phi\nabla_m\phi) - \frac{\phi}{\sqrt{-g}}\partial_m\left(\sqrt{-g}\, g^{mn}\partial_n\phi\right) + m^2\phi^2\right) . \tag{3.17}$$

The second and third terms in the brackets cancel when the equation of motion (3.10) is satisfied, leaving only the total derivative. This we can integrate to give a boundary term on the cutoff surface at $\epsilon = 0$

$$S_\phi = -\frac{1}{2}\int \mathrm{d}^d x\, \sqrt{-\gamma}\, \phi\, n^m \partial_m \phi\big|_{z=\epsilon} \,, \tag{3.18}$$

where the integral is over the x^μ directions, γ is the induced metric on the cutoff surface, and n^m is the-outward pointing unit normal. If we plug in $\phi(\epsilon, x) \approx J(x)\epsilon^{d-\Delta}$ we find

$$S_\phi = \frac{d-\Delta}{2}\frac{L^{d-1}}{\epsilon^{2\Delta-d}}\int \mathrm{d}^d x\, J(x)^2 + \cdots \,, \tag{3.19}$$

where the dots denote subleading terms in the small ϵ expansion. We can cancel this divergence by adding to the action a boundary term of the form

$$S_{\mathrm{ct}} = c\int \mathrm{d}^d x \sqrt{-\gamma}\, \phi^2 \,, \tag{3.20}$$

for some constant c. Again plugging in $\phi(\epsilon, x) \approx J(x)\epsilon^{d-\Delta}$ we find

$$S_{\mathrm{ct}} = \frac{cL^d}{\epsilon^{2\Delta-d}}\int \mathrm{d}^d x\, J(x)^2 + \cdots \,. \tag{3.21}$$

Thus, if we choose $c = (d-\Delta)/(2L)$, the sum $S_\phi + S_{\mathrm{ct}}$ will not have the $1/\epsilon^{2\Delta-d}$ divergence. Typically, for generic values of d and Δ you will need additional counterterms to cancel the subleading divergences in equation (3.19). These counterterms should always be covarint. For scalar fields, this covariance requirement means that the additional counterterms typically involve integrals of $\sqrt{-\gamma}$ times powers of (derivatives of) ϕ. Once all of the divergences are properly cancelled by counterterms, one ends up with a renormalised action with which one can compute correlation functions. One of the exercises will walk you through a concrete correlation function calculation.

The procedure described above is called *holographic renormalisation*, as it is the holographic analog of renormalisation in the dual QFT. Since the divergences that we renormalise in QFT come from the UV (high energies), we should think of the cutoff ϵ as a UV regulator in the QFT. This illustrates an important general property of AdS/CFT, the UV/IR relation. In the gravity side the cutoff ϵ is an IR cutoff, since we are cutting off the part of the spacetime at infinity. This is more transparent in the coordinates (1.35), where a cutoff at small $z = \epsilon$ translates into a cutoff at large radius $r = L^2/\epsilon$. So a UV cutoff in the QFT corresponds to an IR cutoff in the gravity theory. More generally:

UV physics at the boundary corresponds to IR physics in the bulk.

3.5 The metric and the stress tensor

A field that is always present in the bulk gravitational theory is the metric g_{mn}. It turns out that:

The QFT operator dual to the metric is the stress tensor $T_{\mu\nu}$.

The map works as follows. In AdS/CFT we deal with asymptotically-AdS spactimes, which look like AdS at their conformal boundary. Suppose we have a coordinate z, such that the conformal boundary is at $z = 0$. If we choose coordinates such that $g_{zz} = L^2/z^2$ and $g_{mz} = 0$ for $m \neq z$, the metric of a $(d+1)$-dimensional asymptotically-AdS space takes the form

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left[\mathrm{d}z^2 + h_{\mu\nu}(z,x)\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu\right], \tag{3.22}$$

for some tensor $h_{\mu\nu}(z,x)$, where $\mu = 0, 1, \cdots, (d-1)$. Einstein's equations imply that $h_{\mu\nu}(z,x)$ has a near-boundary expansion

$$h_{\mu\nu}(z,x) = h^{(0)}_{\mu\nu}(x)\,(1+\cdots) + h^{(d)}_{\mu\nu}(x) z^d\,(1+\cdots)\,, \tag{3.23}$$

where the coefficients $h^{(0)}_{\mu\nu}(x)$ and $h^{(d)}_{\mu\nu}(x)$ are to be fixed by boundary conditions. The dots denote terms with higher powers of z, with coefficients fixed by the equations of motion. The form of these terms depend on the model in question.

We interpet the leading-order term in the expansion $h^{(0)}_{\mu\nu}(x)$ as the boundary metric — the metric of the manifold on which the dual QFT lives. As for the scalar field, this leading-order term in the near-boundary expansion acts as a source for an operator. Recall that one way to compute correlation functions of the stress tensor $T_{\mu\nu}$ in QFT is to place the QFT on a curved space and take functional derivatives with respect to the metric, so $h^{(0)}_{\mu\nu}(x)$ is indeed the source for the stress tensor. For example,

$$\langle T^{\mu\nu}\rangle = -\frac{2}{\sqrt{-h^{(0)}}}\frac{\delta S^\star_{\text{grav}}}{\delta h^{(0)}_{\mu\nu}}\,. \tag{3.24}$$

Similar to the action for the scalar field in the previous section, the Einstein–Hilbert action diverges due to the $z \to 0$ behaviour of the integrand, so to obtain the correlation functions using this prescription we need to regularise with a cutoff at $z = \epsilon$ and renormalise the divergences with covariant counterterms. However, there is also a boundary term that we need to regardless of the renormalisation issues. The reason is that the Einstein–Hilbert action is incompatible with Dirichlet boundary conditions (i.e. fixing $h^{(0)}_{\mu\nu}$) unless we add the Gibbons–Hawking boundary term to the action:

$$S_{\text{GH}} = -\frac{1}{8\pi G}\int \mathrm{d}^d x\,\sqrt{-\gamma}\,K\,, \tag{3.25}$$

where as before γ_{ab} is the induced metric on the cutoff surface at $z = \epsilon$. The quantity K is the trace of the extrinsic curvature of this surface. Thus, starting from a bulk gravitational action S, the full renormalised gravitational action is

$$S_{\text{grav}} = S + S_{\text{GH}} + S_{\text{ct}}\,, \tag{3.26}$$

for some collection of counterterms that need to be determined on a case-by-case basis. It is the on-shell value of this renormalised action that can be used to compute correlation functions.

4 Holography at finite temperature

We now turn to some concrete holographic calculations of quantities in strongly coupled $\mathcal{N} = 4$ SYM. We will focus on computing some thermodynamic quantities at non-zero temperature, since this provides a relatively simple setting, where we can also straightforwardly compare to perturbative results at weak coupling. For this we will need another entry in the holographic dictionary:

Thermodynamics in the bulk is the same as thermodynamics in the boundary.

For example, the temperature in a holographic QFT is the same as the temperature in the dual gravitational theory. Often this means that there is an black hole in the gravitational solution describing a QFT at non-zero temperature, and we identify the temperature in the QFT as the Hawking temperature. The identification of the partition functions of holographically dual field theories mean that the free energy F in the QFT equals the free energy in the gravitational theory. We can then obtain other thermodynamic quantities by differentiating. For instance, the entropy $S = -\partial F/\partial T$ is the same on both sides. When the gravitational solution contains a black hole, this means that the entropy in the QFT is given by the Bekenstein–Hawking entropy.

A five-dimensional action. The action for type IIB supergravity is quite complicated. Fortunately, for the solutions that we'll consider the only non-trivial fields are the metric and the four-form gauge field $C_{(4)}$, which has field strength $F_{(5)}$. The action contains terms involving these fields,

$$S = \frac{1}{16\pi G_{(10)}} \int \mathrm{d}^{10}x \sqrt{-g} \left(R - \frac{1}{4 \times 5!} F_5^2 + \cdots \right) , \tag{4.1}$$

where the dots denote terms involving other fields. Moreover, in the solutions we'll consider the S^5 plays only a spectator role, since all of the fields will be constant on it. We can then integrate over the S^5. The upshot is that we can get away with using the Einstein–Hilbert action with a cosmological constant $\Lambda = -6/L^2$ arising from the $F_{(5)}^2$ term in the action, namely

$$S_{\mathrm{EH}} = \frac{1}{16\pi G_{(5)}} \int \mathrm{d}^5x \sqrt{-g} \left(R + \frac{12}{L^2} \right) . \tag{4.2}$$

The five-dimensional Newton constant $G_{(5)}$ is related to the ten-dimensional one by a factor of the volume of the five-sphere,

$$\frac{1}{G_{(5)}} = \frac{\mathrm{vol}(S^5)}{G_{(10)}} = \frac{\pi^3 L^5}{G_{(10)}} . \tag{4.3}$$

Using the explicit value of $G_{(10)}$ from equation (3.4), we arrive at the formula

$$\frac{1}{G_{(5)}} = \frac{2N^2}{\pi L^3} . \tag{4.4}$$

We will use this relation very frequently to relate thermodynamic quantities in type IIB supergravity and in $\mathcal{N} = 4$ SYM.

4.1 Holographic calculation of free energy

The five-dimensional AdS-Schwarzschild solution is

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left(-u(z)\,\mathrm{d}t^2 + \mathrm{d}\vec{x}^{\,2} + \frac{\mathrm{d}z^2}{u(z)}\right), \qquad u(z) = 1 - \frac{z^4}{z_H^4}\,, \tag{4.5}$$

where $\vec{x} = (x^1, x^2, x^3)$, and $\mathrm{d}\vec{x}^{\,2}$ denotes the Cartesian metric in $\mathbb{R}^3$. This solution has an event horizon at $z = z_H$ where $u(z_H) = 0$. To study thermodynamics, we Wick rotate to imaginary time $\tau = it$, in terms of which the metric becomes

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left(u(z)\,\mathrm{d}\tau^2 + \mathrm{d}\vec{x}^{\,2} + \frac{\mathrm{d}z^2}{u(z)}\right), \qquad u(z) = 1 - \frac{z^4}{z_H^4}\,, \tag{4.6}$$

We can then compute the Hawking temperature using the procedure outlined in section 1.5

$$T = \frac{|u'(z_H)|}{4\pi} = \frac{1}{\pi z_H}\,. \tag{4.7}$$

The holographic dictionary tells us that this is the temperature of the dual QFT.

Similarly, the entropy S in the QFT is given holographically by the the Bekenstein–Hawking entropy S of the black hole. This yields

$$S = \frac{L^3 V}{4G_{(5)} z_H^3}\,, \qquad V \equiv \int \mathrm{d}^3 x\,. \tag{4.8}$$

Recall that the dual $\mathcal{N} = 4$ SYM theory lives in Minkowski space with cartesian spatial coordinates $\vec{x}$. The factor of V is therefore the volume of the QFT system. If we integrate over the whole of $\mathbb{R}^3$ then this volume is infinite. We should therefore imagine that we place our system in finite volume by restricting the range of the $\vec{x}$ directions. We can then divide out by V to obtain densities that make sense in the $V \to \infty$ limit. For example, from the coefficient in equation (4.8) we read off that the entropy density $s = S/V$ is

$$s = \frac{L^3}{4G_{(5)} z_H^3} = \frac{\pi^2 N^2 T^3}{2}\,. \tag{4.9}$$

In the second equality we used equations (4.4) and (4.7) to replace the gravitational quantities L, $G_{(5)}$, and z_H with quantities that make sense in the QFT, namely N and T.

The fact that we found $s \propto T^3$ in equation (4.9) was guaranteed by dimensional analysis. Since $\mathcal{N} = 4$ SYM is a conformal field theory, it contains no dimensionful parameters. Thus, the only dimensionful quantity for things to depend on is the temperature, which in natural units has dimension [mass]. Since entropy is dimensionless, entropy *density* has dimensions of $[\text{mass}]^3$, and hence $s \propto T^3$. The fact that $s \propto N^2$ has a natural interpretation, the number of degrees of freedom of $\mathrm{SU}(N)$ $\mathcal{N} = 4$ SYM is proportional to N^2 (at large N), as we discuss further below.

Now let us compute the (Helmholtz) free energy F of strongly coupled planar $\mathcal{N} = 4$ SYM as a function of temperature. Recall that the holographic dictionary equates the partition functions of a QFT and its gravitational dual. Since the partition function and the free energy are related by $Z = \exp(-F/T)$, in the semiclassical limit where $Z_{\rm grav} \approx e^{-I^\star_{\rm grav}}$ we have that the free energy is given by

$$F = T I^\star_{\rm grav}\,. \tag{4.10}$$

Our task is therefore to evaluate the Euclidean signature gravitational action $I_{\rm grav}$, on the solution (4.6).

The action has several ingredients. The first is one that we have already discussed, the Einstein–Hilbert piece

$$I_{\rm EH} = -\frac{1}{16\pi G_{(5)}} \int \mathrm{d}^5x\, \sqrt{\det g} \left(R + \frac{12}{L^2}\right). \tag{4.11}$$

The determinant of the induced metric in equation (4.6) and its Ricci scalar are

$$\det g = \frac{L^{10}}{z^{10}}\,, \qquad R = -\frac{20}{L^2}\,. \tag{4.12}$$

Plugging these in, we find that the on-shell Einstein–Hilbert action is

$$\begin{aligned} I^\star_{\rm EH} &= \frac{L^3}{2\pi G_{(5)}} \int_0^\beta \mathrm{d}\tau \int_\epsilon^{z_H} \frac{\mathrm{d}z}{z^5} \int \mathrm{d}^3x \\ &= \frac{N^2\beta V}{4\pi^2}\left(\frac{1}{\epsilon^4} - \frac{1}{z_H^4}\right). \end{aligned} \tag{4.13}$$

The integrand diverges at $z \to 0$ (i.e. close to the asymptotically-AdS boundary), so we have regulated this divergence by introducing a small near-boundary cutoff ϵ.

The next ingredient in the action is the Gibbons–Hawking–York (GHY) boundary term evaluated on the cutoff surface at $z = \epsilon$. We use y and γ to denote the coordinates and induced metric on the cutoff surface, respectively, and K to denote its mean curvature. The GHY term is

$$I_{\rm GHY} = -\frac{1}{8\pi G_{(5)}} \int \mathrm{d}^4y \sqrt{\det\gamma}\, K\,. \tag{4.14}$$

The natural choice of coordinates on the cutoff surface is $y = (\tau, x^1, x^2, x^3)$ is

$$\gamma_{ij}\,\mathrm{d}y^i\,\mathrm{d}y^j = \frac{L^2}{\epsilon^2}\left(-u(\epsilon)\,\mathrm{d}\tau^2 + \mathrm{d}\vec{x}^{\,2}\right), \tag{4.15}$$

and one finds that

$$\sqrt{\det\gamma}\,K = 2L^3\left(\frac{2}{\epsilon^4} - \frac{1}{z_H^4}\right). \tag{4.16}$$

Substituting this into the GHY action and using equation (4.4) to rewrite $G_{(5)}$ in term of N, we arrive at

$$I^\star_{\rm GHY} = -\frac{N^2\beta V}{2\pi^2}\left(\frac{2}{\epsilon^4} - \frac{1}{z_H^4}\right). \tag{4.17}$$

Summing equations (4.13) and (4.17) we obtain

$$I^\star_{\rm EH} + I^\star_{\rm GHY} = \frac{N^2\beta V}{4\pi^2}\left(-\frac{3}{\epsilon^4} + \frac{1}{z_H^4}\right). \tag{4.18}$$

This can't be the full story for the on-shell action since it diverges when we remove the cutoff by sending $\epsilon \to 0$. We need to remove this divergence through holographic renormalisation. As in the example with the scalar field, the holographic renormalisation procedure works by writing down covariant counterterms evaluated on the cutoff surface at $z = \epsilon$, and choosing the coefficients such that they cancel all divergences in the on-shell action. In our current example, where the only field is the metric, the only ingredient we have with which to build covariant counterterms is the induced metric γ_{ij} on the cutoff surface. Since $\sqrt{\det \gamma}$ diverges as ϵ^{-4} when $\epsilon \to 0$, a counterterm of the form

$$I_{\text{ct}} = \frac{c}{8\pi G_{(5)} L} \int \mathrm{d}^4 y \, \sqrt{\det \gamma} \tag{4.19}$$

should do the trick to cancel the ϵ^{-4} divergence in the $I^\star_{\text{EH}} + I^\star_{\text{GHY}}$, once we figure out the correct coefficient c.[4] Evaluating I_{ct} we find

$$I^\star_{\text{ct}} = c\frac{N^2 \beta V}{4\pi^2}\left(\frac{1}{\epsilon^4} - \frac{1}{2z_H^4}\right) + O(\epsilon^4). \tag{4.20}$$

Comparing to equation (4.18), we see that if we choose $c = 3$ then $I^\star_{\text{grav}} \equiv I^\star_{\text{EH}} + I^\star_{\text{GHY}} + I^\star_{\text{ct}}$ will be finite in the limit $\epsilon \to 0$. Thus we should take the counterterm action to be

$$I_{\text{ct}} = \frac{3}{8\pi G_{(5)} L} \int \mathrm{d}^4 y \, \sqrt{\det \gamma} \tag{4.21}$$

In total, the on-shell action is therefore

$$I^\star_{\text{grav}} = -\frac{N^2 \beta V}{8\pi^2 z_H^4} = -\frac{\pi^2 N^2 V T^3}{8}\,. \tag{4.22}$$

Finally, using $F = T I^\star$, we obtain the free energy density $f = F/V$,

$$f = -\frac{\pi^2 N^2 T^4}{8}\,. \tag{4.23}$$

As a check of our calculation, the entropy density computed from the thermodynamic relation $s = -\partial f/\partial T$ matches the Bekenstein–Hawking entropy in equation (4.9).

4.2 Comparison to weak coupling

A single free, massless scalar field at temperature T has free energy density $f = -\pi^2 T^4/90$. A free massless vector field has two polarisations and therefore has free energy density twice that of a scalar, i.e. $f = -\pi^2 T^4/45$. Finally, a free massless Dirac fermion has free energy density $f = -7\pi^2 T^4/180$. In total, then, a free theory consisting of n_s massless scalars, n_f massless fermions, and n_v massless vector fields, has free energy density

$$f = -\frac{2n_s + 7n_f + 4n_v}{180}\pi^2 T^4\,. \tag{4.24}$$

[4] We included a factor of L in the denominator of the coefficient of the counterterm to make the coefficient c dimensionless.

In $\mathcal{N}=4$ SYM there are six scalars, two fermions, and one vector, that all transform in the adjoint representation of SU(N) which is N^2-1 dimensional. Thus, $n_s = 6(N^2-1)$, $n_f = 2(N^2-1)$ and $n_v = N^2-1$. Plugging these numbers into equation (4.24) we find

$$f = -\frac{\pi^2(N^2-1)T^4}{6} \approx -\frac{\pi^2 N^2 T^4}{6}\,, \tag{4.25}$$

where in the second equality we have taken the large-N limit.

Comparing to equation (4.23), we see that the large-N free energies at $\lambda = 0$ and $\lambda \to \infty$ take a very similar form, just with different numerical coefficients. The free field result demonstrates why it is natural that f in equation (4.23) is proportional to N^2, since the number of degrees of freedom of $\mathcal{N}=4$ SYM is proportional to N^2 at large N.

The fact that the numerical coefficients in equations (4.23) and (4.25) are different is not surprising, since they correspond to different regimes in the coupling constant λ

$$f = -\frac{\pi^2 N^2 T^4}{6} g(\lambda) \tag{4.26}$$

for some function $g(\lambda)$. The free field results tells us that $g(0)=1$, while the holographic result tells us that $\lim_{\lambda\to\infty} g(\lambda) = 3/4$. It is possible to improve on these limits and compute corrections. At weak coupling $\lambda \ll 1$, the free energy of $\mathcal{N}=4$ SYM has been computed in perturbation theory up to $O(\lambda^2)$, while holography has been used to compute the leading-order, $O(\lambda^{-3/2})$, correction to $g(\lambda)$ at large but finite coupling. These corrections are[5]

$$g(\lambda) = \begin{cases} 1 - \frac{3\lambda}{2\pi^2} + (3+\sqrt{2})\left(\frac{\lambda}{\pi^2}\right)^{3/2} + (-5.97... + \frac{3}{2}\log\lambda)\lambda^2 + \cdots\,, & \lambda \ll 1\,, \\ \frac{3}{4} + \frac{45}{32}\zeta(3)\lambda^{-3/2} + \cdots\,, & \lambda \gg 1\,, \end{cases} \tag{4.27}$$

where ζ is the Riemann zeta function. We plot both expressions for $g(\lambda)$ in figure 4. The finite λ corrections push $g(\lambda)$ in the right direction for intermediate values of the coupling: the perturbative corrections tend to make $g(\lambda) < 1$ at values of λ for which they are reliable, while the $O(\lambda^{-3/2})$ correction to the holographic result makes $g(\lambda) > 3/4$. Unfortunately the regimes in which we can trust the two expansions do not overlap, so we cannot make a more precise comparison between the perturbative and holographic results.

4.3 The Hawking–Page phase transition

4.3.1 Thermal AdS

The metric of global AdS was written in equation (1.42). For $d=4$ it is[6]

$$\mathrm{d}s^2 = \left(1 + \frac{r^2}{L^2}\right)\mathrm{d}\tau^2 + \frac{\mathrm{d}r^2}{1+r^2/L^2} + r^2\,\mathrm{d}\Omega_3^2\,, \tag{4.28}$$

where $\mathrm{d}\Omega_3^2$ denotes the metric on a unit-radius S^3. Using three angular coordinates $\theta \in [0,\pi]$, $\phi \in [0,\pi]$, and $\psi \in [0,2\pi]$, we can write

$$\mathrm{d}\Omega_3^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2 + \sin^2\theta\sin^2\phi\,\mathrm{d}\psi^2\,. \tag{4.29}$$

[5]In equation (4.27), 5.97... are the first few decimal digits of a number which is known exactly in terms of logs and the Riemann zeta function, but rather lengthy to write out. The exact expression may be found in ref. [2].

[6]In this section we call the radial coordinate r instead of ρ since it is a little easier to write.

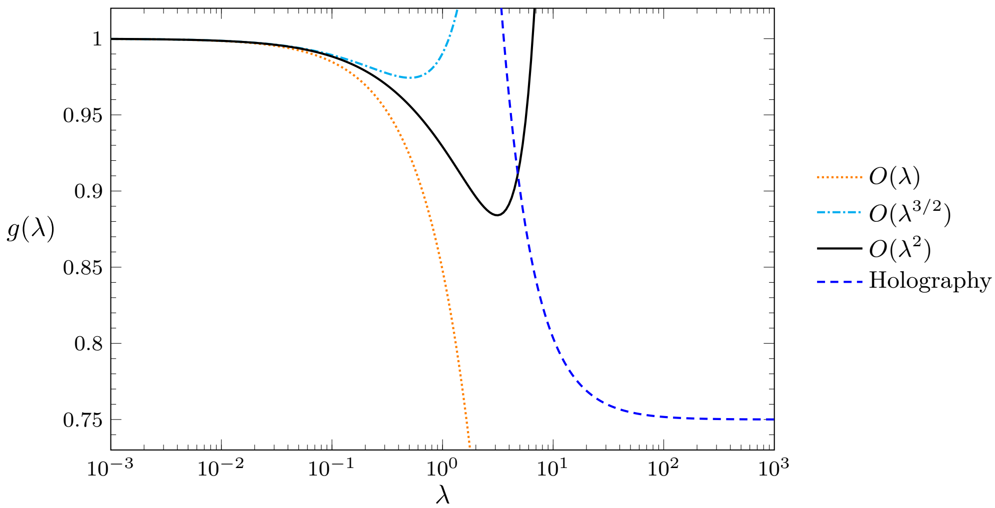


Figure 4: The function $g(\lambda)$ that determines the coupling-constant dependence of the free energy of $\mathcal{N}=4$ SYM. The curves marked $O(\lambda)$, $O(\lambda^{3/2})$, and $O(\lambda^2)$ show the perturbative result up to the indicated order in λ. The dashed blue curve is the result from holography up to $O(\lambda^{-3/2})$. Figure adapted from ref. [2].

The determinant of the metric in equation (4.28) is $\det g = r^6 \sin^4\theta \sin^2\phi$, while the Ricci scalar is

$$R = -\frac{20}{L^2}\,, \tag{4.30}$$

as appropriate for AdS_5 with curvature radius L. Consequently, the Einstein–Hilbert action evaluated on this solution is

$$\begin{aligned} I^{\star}_{\mathrm{EH}} &= -\frac{1}{16\pi G_{(5)}} \int \mathrm{d}^5x\,\sqrt{\det g}\left(R + \frac{12}{L^2}\right) \\ &= \frac{1}{2\pi G_{(5)} L^2} \int_0^{\beta} \mathrm{d}\tau \int_0^{L^2/\epsilon} \mathrm{d}r\, r^3 \int_0^{\pi} \mathrm{d}\theta \sin^2\theta \int_0^{\pi} \mathrm{d}\phi \sin\phi \int_0^{2\pi} \mathrm{d}\psi\,. \end{aligned} \tag{4.31}$$

As before, the r integral would have diverged if we had integrated all the way to the boundary of AdS at $r=\infty$, so we have integrated only up to some large $r = L^2/\epsilon$. Explicitly performing all the integrals and using equation (4.4) to express $G_{(5)}$ in terms of N, we find

$$I^{\star}_{\mathrm{EH}} = \frac{N^2 L^3 \beta}{2}\frac{1}{\epsilon^4}\,. \tag{4.32}$$

The action also receives a contribution from the Gibbons–Hawking–York term at the cutoff surface. The induced metric on the cutoff surface is given by

$$\gamma_{ij}\,\mathrm{d}y^i\,\mathrm{d}y^j = \left(1 + \frac{L^2}{\epsilon^2}\right)\mathrm{d}\tau^2 + \frac{L^4}{\epsilon^2}\,\mathrm{d}\Omega_3^2\,, \tag{4.33}$$

so that its determinant and the mean curvature are

$$\det\gamma = \frac{L^{12}}{\epsilon^8}(L^2+\epsilon^2)\sin^4\theta\sin^2\phi \qquad K = \frac{4L^2 + 3\epsilon^2}{L^2\sqrt{L^2+\epsilon^2}}\,. \tag{4.34}$$

Thus, the GHY term is

$$
\begin{aligned}
I^{\star}_{\mathrm{GHY}} &= -\frac{1}{8\pi G_{(5)}} \int_{r=L^2/\epsilon} \mathrm{d}^4 y \, \sqrt{\det \gamma}\, K \\
&= N^2 \beta \left(-\frac{2L^3}{\epsilon^4} - \frac{3L}{2\epsilon^2} \right) .
\end{aligned} \tag{4.35}
$$

Adding this to the Einstein-Hilbert action we find

$$
I^{\star}_{\mathrm{EH}} + I^{\star}_{\mathrm{GHY}} = -\frac{3N^2\beta}{2L} \left(\frac{L^4}{\epsilon^4} + \frac{L^2}{\epsilon^2} \right) . \tag{4.36}
$$

As before, the divergences in this on-shell action should be cancelled by the addition of counterterms. There are two counterterms that we can write

$$
\begin{aligned}
I_{\mathrm{ct}} &= \frac{1}{8\pi G_{(5)}} \int \mathrm{d}^4 y \, \sqrt{\det \gamma} \left(\frac{3}{L} + \frac{1}{4L} \mathcal{R}[\gamma] \right) \\
&= \frac{3N^2\beta}{2L} \left(\frac{L^4}{\epsilon^4} + \frac{L^2}{\epsilon^2} + \frac{1}{8} \right) + O(\epsilon^4)
\end{aligned} \tag{4.37}
$$

Thus, the free energy $F = T(I^{\star}_{\mathrm{EH}} + I^{\star}_{\mathrm{GHY}} + I_{\mathrm{ct}})$ is given by

$$
F = \frac{3N^2}{16L} . \tag{4.38}
$$

The free energy in this phase is independent of temperature and hence the thermodynamic entropy, $S = -\partial F / \partial T$, vanishes. Consequently, the total energy of the system, $E = F + TS$, is equal to the free energy, so we have

$$
E = \frac{3N^2}{16L} . \tag{4.39}
$$

4.3.2 Black hole phase

The gravitational solution that competes with the one in the previous section is

$$
\mathrm{d}s^2 = f(r)\, \mathrm{d}\tau^2 + \frac{r^2}{f(r)} + r^2\, \mathrm{d}\Omega_3^2 , \qquad f(r) = \frac{(r^2 - r_H^2)(r^2 + r_H^2 + L^2)}{L^2 r^2} , \tag{4.40}
$$

where r_H is a constant. This solution describes a black hole in asymptotically global AdS_5, with event horizon at $r = r_H$, where $f(r_H) = 0$. Notice that the solution in equation (4.40) reduces to that in equation (4.28) for $r_H = 0$.

The Hawking temperature of the black hole is

$$
T = \frac{f'(r_H)}{4\pi} = \frac{L^2 + 2r_H^2}{2\pi L^2 r_H} . \tag{4.41}
$$

The Hawking temperature is plotted against r_H in figure 5. Unlike the Schwarzschild black hole in the Poincaré patch of AdS, this family of black holes cannot reach all the way to $T = 0$. Instead, the black holes reach a minimum temperature at $r_H = L/\sqrt{2}$, where $T = T_{\min} = \sqrt{2}/\pi L$. Above $T_{\min}$ there are two values of r_H for every T. Black

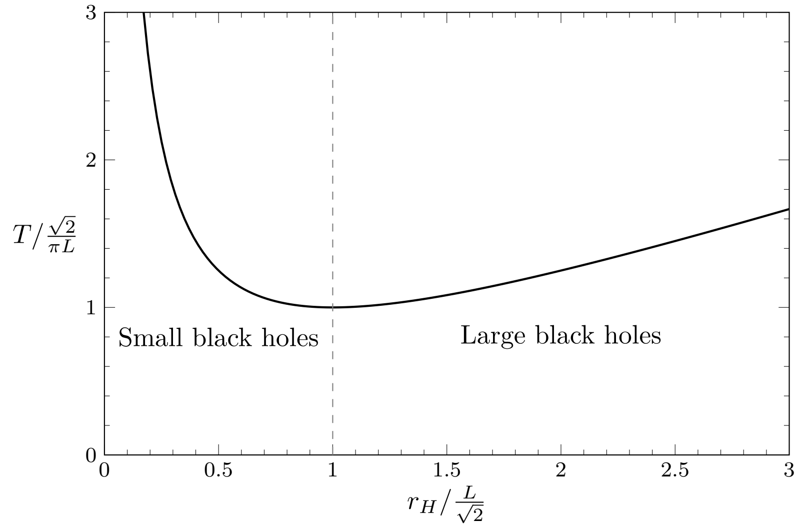


Figure 5: Temperature versus horizon location for the black hole solution in equation (4.40).

holes with $r_H < L/\sqrt{2}$ are referred to as *small black holes*, while those with $r_H > L/\sqrt{2}$ are referred to as *large black holes*. We can solve equation (4.41) to find that the small and the large black hole at temperature T have radii given by

$$r_H = \frac{L}{2}\left(\pi LT \pm \sqrt{\pi^2 L^2 T^2 - 2}\right), \tag{4.42}$$

where the plus sign is for large black holes and the minus sign for small black holes.

If we are looking for solutions that can describe $\mathcal{N} = 4$ SYM at $T < T_{\min}$, the only option is the thermal AdS solution described in the previous section. On the other hand, if we are looking for solutions to describe $\mathcal{N} = 4$ SYM at $T > T_{\min}$, there are three options: thermal AdS, a small black hole, or a large black hole. To find out which one dominates we need to compare their free energies.

The computation of the free energy of the black hole solutions proceeds almost identically to that for thermal AdS, described in the previous subsection. The Ricci scalar of the solution in equation (4.40) is $R = -20/L^2$, the same as for thermal AdS. The determinant of the metric is also the same. Thus, in computing $I^\star_{\text{EH}}$, the only change from equation (4.31) is that the lower limit on the r integration is now r_H instead of zero,

$$\begin{aligned} I^\star_{\text{EH}} &= \frac{1}{2\pi G_{(5)} L^2}\int_0^\beta \mathrm{d}\tau \int_{r_H}^{L^2/\epsilon} \mathrm{d}r\, r^3 \int_0^\pi \mathrm{d}\theta \sin^2\theta \int_0^\pi \mathrm{d}\phi \sin\phi \int_0^{2\pi}\mathrm{d}\psi \\ &= \frac{N^2\beta}{2L}\left(\frac{L^4}{\epsilon^4} - \frac{r_H^4}{L^4}\right). \end{aligned} \tag{4.43}$$

Computing the Gibbons-Hawking term and the counterterms, we find

$$\begin{aligned} I^\star_{\text{GH}} &= \frac{2N^2\beta}{L}\left(-\frac{L^4}{\epsilon^4} - \frac{3L^2}{4\epsilon^2} + \frac{r_H^2}{2L^2} + \frac{r_H^4}{2L^4}\right), \\ I^\star_{\text{ct}} &= \frac{3N^2\beta}{2L}\left(\frac{L^4}{\epsilon^4} + \frac{L^4}{\epsilon^2} + \frac{1}{8} - \frac{r_H^2}{2L^2} - \frac{r_H^4}{2L^4}\right). \end{aligned} \tag{4.44}$$

Putting all of this together, we find for the free energy $F = T(I^\star_{\rm EH} + I^\star_{\rm GH} + I^\star_{\rm ct})$,

$$F = \frac{N^2}{16L^2}\left(3 + \frac{4r_H^2}{L^2} - \frac{4r_H^4}{L^4}\right). \tag{4.45}$$

Finally, we would like to express this completely in terms of the temperature. Using equation (4.42) to eliminate r_H, we find

$$F_\pm = \frac{\pi N^2 T}{8}\left[\pi LT\left(3 - \pi^2L^2T^2\right) \mp \left(\pi^2L^2T^2 - 2\right)^{3/2}\right]. \tag{4.46}$$

Notice that $F_+ \leq F_-$ for all T, with equality only at $T = T_{\rm min}$ where the two branches of solutions meet. Thus, large black holes are always thermodynamically preferred over small black holes. What remains is to compare the free energy of large black holes to that of the thermal gas.

At large temperatures, $TL \gg 1$, we find

$$F_+ \approx -\frac{\pi^3 N^2}{4}L^3T^4. \tag{4.47}$$

Since this is negative, it is smaller than the postive free energy $F_{\rm TG} = 3N^2/16L$ for the thermal gas given in equation (4.38). Consequently, large black holes are preferred over the thermal gas at large temperatures. On the other hand, at $T = T_{\rm min} = \sqrt{2}/\pi L$ we find $F_+ = N^2/4L > F_{\rm TG}$. Thus the thermal gas is preferred at $T = T_{\rm min}$. There must therefore be a phase transition at some critical $T = T_c > T_{\rm min}$, where the two free energies are equal. Solving $F_+ = F_{\rm TG}$ for T, we find the critical temperature to be just slightly larger than $T_{\rm min}$,

$$T_c = \frac{3}{2\pi L} \approx 1.06\, T_{\rm min}. \tag{4.48}$$

We plot the free energies of the different types of solution in figure 6a.

The Bekenstein–Hawking entropy of the black hole solution (4.40) is

$$S = \frac{{\rm Vol}(S^3) r_H^3}{4G_{(5)}} = \frac{\pi N^2 r_H^3}{L^3}. \tag{4.49}$$

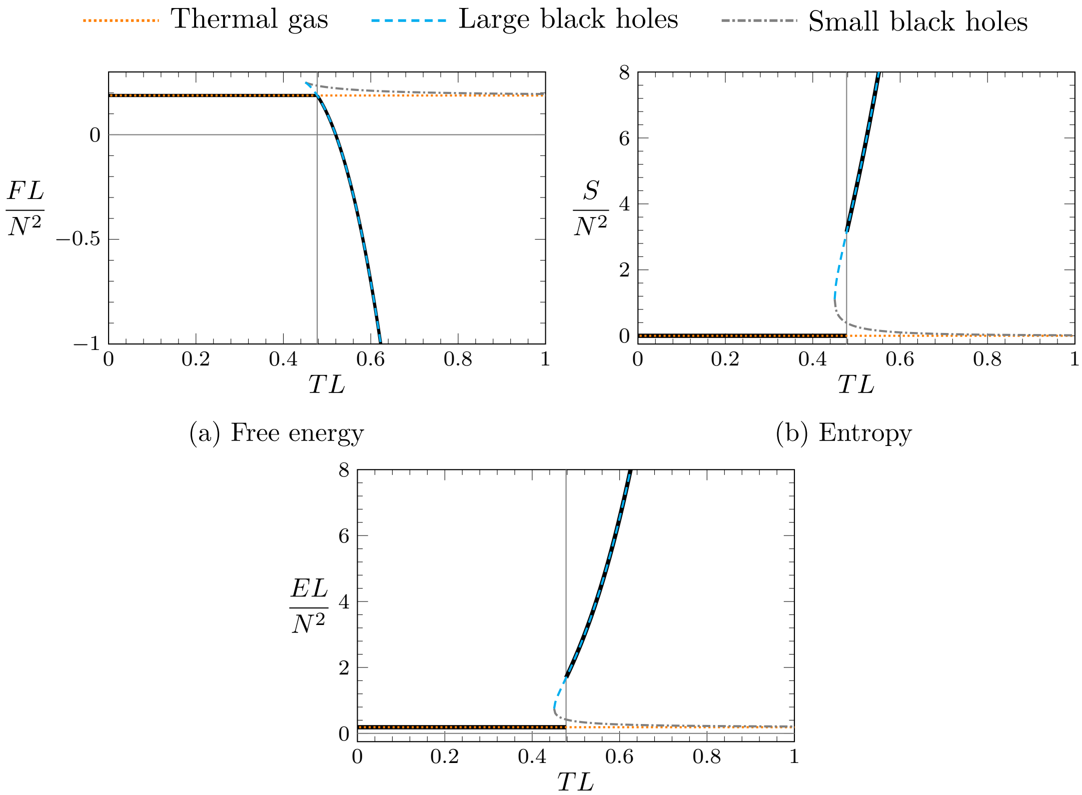
Using equation (4.42) to replace r_H with T, we find that the entropies large and small black holes at temperature T, S_+ and S_- respectively, are given by

$$S_\pm = \frac{\pi N^2}{8}\left(\pi LT \pm \sqrt{\pi^2L^2T^2 - 2}\right)^3 \tag{4.50}$$

As a cross check, this should match the thermal entropy obtained by differentiating the free energy with respect to temperature, and indeed it does; it is straightforward to check that $S_\pm = -\partial F_\pm/\partial T$. Knowing the free energy and entropy, we can extract the energy via the thermodynamical relation $E = F + TS$, which yields

$$E_\pm = \frac{3\pi^2N^2LT^2}{8}\left(\pi^2L^2T^2 - 1 \pm \pi LT\sqrt{\pi^2L^2T^2 - 2}\right), \tag{4.51}$$

for large and small black holes, respectively. The entropy and energy as functions of temperature are plotted in figures 6b and 6c.



(a) Free energy

(b) Entropy

(c) Energy

Figure 6: The free energy, entropy, and energy of strongly coupled, planar $\mathcal{N} = 4$ super Yang–Mills with gauge group $\mathrm{SU}(N)$, on a three-sphere of radius L as a function of temperature T. The thick black curves correspond to the thermodynamically dominant phase. The other curves, with various coulours and dashings, indicate the values of these thermodynamic quantities in each of the gravitational phases.

4.4 A test of AdS/CFT

The result in equation (4.39) provides us with an opportunity to test the AdS/CFT conjecture, as follows.

Every four-dimensional CFT has a number a known as "the type A Weyl anomaly coefficient", which controls various physical quantities. An important property of this number is that it is independent of any marginal couplings that the theory may have. Since the 't Hooft coupling λ is a marginal coupling, this means that if we can compute a using holography (i.e. at $\lambda \to \infty$), the result should match the result at $\lambda = 0$ where the theory is free.

It turns out that one of the physical quantities controlled by a is the Casimir energy the four-dimensional CFT at zero temperature on $\mathbb{R} \times \mathrm{S}^3$. The relation is

$$E = \frac{3a}{4L}\,, \tag{4.52}$$

where L is the radius of S^3.[7] This allows us to read off from equation (4.39) the type A Weyl anomaly coefficient of $\mathcal{N}=4$ SYM with gauge group $\mathrm{SU}(N)$ at $N \gg 1$,

$$a = \frac{N^2}{4}\,. \tag{4.53}$$

Let's compare to $\lambda = 0$, where free field methods may be used to compute a. For a free conformal theory consisting of n_s massless real scalars, n_f massless Dirac fermions, and n_v massless vector fields, the type A Weyl anomaly coefficient is

$$a = \frac{n_s + 11 n_f + 62 n_v}{360}\,, \tag{4.54}$$

For the case of $\mathcal{N}=4$ SYM, there are six real scalar fields that each transform in the adjoint representation of $\mathrm{SU}(N)$. Since the adjoint representation is (N^2-1)-dimensional this adds up to $n_s = 6(N^2-1)$. Similarly, there are two Dirac fermions and a single vector field, each transforming in the adjoint representation, so $n_f = 2(N^2-1)$ and $n_v = N^2-1$. Plugging the numbers into equation (4.54), one finds

$$a = \frac{N^2-1}{4} \approx \frac{N^2}{4}\,, \tag{4.55}$$

where in the second equality we have taken the large-N limit and kept only the leading term. The right-hand side of equation (4.55) exactly matches the holographic result in equation (4.53)!

5 Wilson lines

5.1 Wilson lines in gauge theory

An important family of operators in gauge theory are Wilson lines. To build these operators we first note that since the gauge field A_μ has a single lower index (in the language of differential geometry it is a one-form) we can integrate it along curve C in spacetime. A Wilson line operator $U[C]$ is defined by exponentiating this integral,

$$U[C] = \mathcal{P}\exp\left(i\int_C \mathrm{d}x^\mu A_\mu\right), \tag{5.1}$$

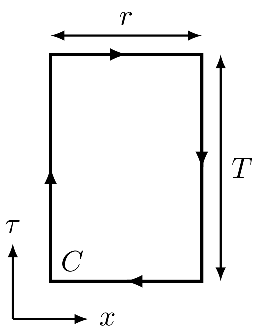
where $\mathcal{P}$ denotes that this is a *path-ordered exponential.* Path ordering is like the time ordering encountered in the Dyson series expression for the time evolution operator. Concretely, suppose we paramaterise the curve C by a coordinate $\tau \in [\tau_i, \tau_f]$, so that

$$\int \mathrm{d}x^\mu A_\mu = \int_{\tau_i}^{\tau_f} \mathrm{d}\tau\, \dot{x}^\mu A_\mu\,, \tag{5.2}$$

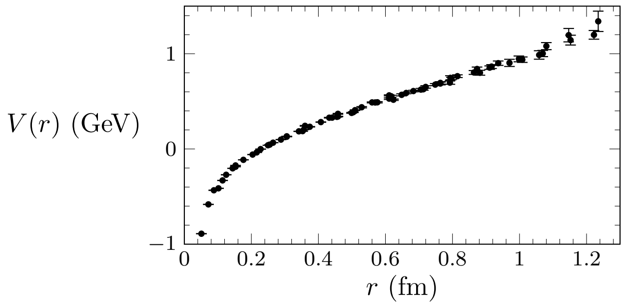
where the dot denotes differentiation with respect to τ. Then

$$\mathcal{P}\exp\left(i\int_C \mathrm{d}x^\mu A_\mu\right) = 1 + i\int_{\tau_i}^{\tau_f}\mathrm{d}\tau\,\dot{x}^\mu A_\mu - \int_{\tau_i}^{\tau_f}\mathrm{d}\tau_1\int_{\tau_i}^{\tau_1}\mathrm{d}\tau_2\,\dot{x}^\mu(\tau_1)A_\mu(\tau_1)\dot{x}^\mu(\tau_2)A_\mu(\tau_2) + \cdots \tag{5.3}$$

[7] If you have studied two-dimensional conformal field theory you will likely have encountered the central charge c, which in 2D plays an analogous role to a in 4D. There is similar result to equation (4.52) for a two-dimensional conformal field theory on $\mathbb{R}\times\mathrm{S}^1$, $E = -c/12L$.



(a) Rectangular Wilson loop.



(b) Quark-antiquark potential in SU(3) Yang–Mills.

Figure 7: **(a):** The rectangular Wilson loop used to diagnose confinement. **(b):** Lattice results for the quark-antiquark potential in SU(3) theory, exhibiting linear growth at large r. Data taken from ref. [3].

Wilson lines are important because they represent the heavy test charges. To see this, suppose we want to compute the expectation value the operator in equation (5.1) using the path integral formalism. The path integral naturally implements the path ordering, so that

$$\langle U[C]\rangle = \int \mathcal{D}A \exp\left(iS + i\int_C \mathrm{d}x^\mu A_\mu\right). \tag{5.4}$$

This looks like the partition function for the theory obtained by adding a term to the action consisting of the gauge field integrated along the curve C. But this is precisely the action for a test charge following this curve. We will refer to such a charge as a "quark", in analogy to QCD.

The Wilson line operator is not gauge invariant, but we can obtain something gauge invariant if take C to be a closed loop and trace over the gauge indices. This gives an gauge-invariant operator called a Wilson loop $W[C]$,

$$W[C] = \operatorname{tr}\mathcal{P}\exp\left(i\oint_C \mathrm{d}x^\mu A_\mu\right). \tag{5.5}$$

One of the most important uses of the Wilson loop is to measure confinement. Suppose we go to Euclidean signature and take the closed loop C to be a rectangle in the (τ, x) plane, with sides of length T and r, as depicted in figure 7a, where we take T to be very large. The Wilson loop then behaves as

$$\langle W[C]\rangle \sim e^{-TV(r)}, \tag{5.6}$$

where $V(r)$ is the potential energy for a quark and an antiquark separated by a distance r. Roughly, this relation arises because at late Euclidean times correlation functions behave as $e^{-E_0 T}$, where E_0 is the lowest energy state. The Wilson loop corresponds to a quark and an antiquark propagating for a time T, separated by a distance r. The energy of this state is precisely what we mean by the potential energy.

There are different possibilities for how the potential $V(r)$ behaves at large r. In QED one finds the Coulomb potential $V(r) \propto 1/r$. On the other hand, the characteristic of a confining theory is that the potential grows linearly in r at large distances, $V(r) \approx \sigma r$ for some constant σ known as the string tension. For example, the quark-antiquark potential in in SU(3) Yang–Mills theory, computed using lattice gauge theory [3], is plotted in figure 7b, where one can see the linear growth of the potential at large r. Thus, in a confining theory the rectangular Wilson loop behaves for large r as

$$\langle W[C]\rangle \sim e^{-\sigma T r} = e^{-\sigma A}\,, \tag{5.7}$$

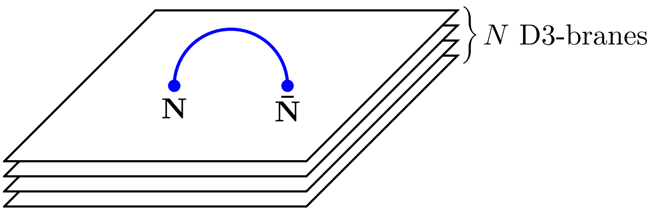
where $A = Tr$ is the area of the loop.

5.2 Wilson loops from holography

It's time for another entry in the holographic dictionary:

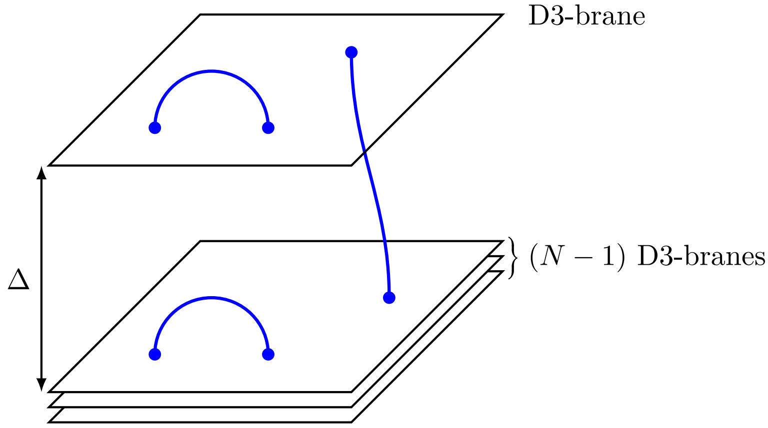
> The expectation value of a Wilson loop $\langle W[C]\rangle$ in $\mathcal{N} = 4$ SYM is given holographically by the on-shell action of a string in $\mathrm{AdS}_5 \times \mathrm{S}^5$ that reaches the boundary alon the curve C.

To motivate the description of Wilson lines in holography, recall that the AdS/CFT conjecture arose from considering a stack of D3-branes. Suppose we have a stack of N D3-branes, corresponding to gauge group SU(N). The adjoint-valued fields of $\mathcal{N} = 4$ SYM arise from strings with both endpoints on the stack. Crucially for this, the endpoints of the string transform in either the fundamental $\mathbf{N}$ or antifundamental representation $\bar{\mathbf{N}}$ of SU(N), and we obtain the adjoint representation because it appears in the relation $\mathbf{N} \otimes \bar{\mathbf{N}} = (\text{adjoint}) + (\text{singlet})$. Here is a cartoon:



The parallelograms represent the stack of D3-branes. Although they are drawn separated, this is only to make them visually distinct. You should think of them as on top of each other. The blue arc represents a string with both ends on the D3-brane stacks. We have indicated the representations of SU(N) that the endpoints transform under.

Now take one of the D3-branes and separate it from the others by a distance Δ. This spontaneously breaks the SU(N) symmetry to SU($N-1$)$\times$U(1). Because of the separation we now have different types of strings, as depicted in the following cartoon:



As before there are strings that begin and end on the stack. They correspond to the $\mathrm{SU}(N-1)$ gauge bosons. There are also strings that begin and end on the separated D3-brane, corresponding to the gauge bosons of the $\mathrm{U}(1)$ factor.

Finally, there are strings with one end on the stack and one end on the separated brane. They transform in the fundamental or antifundamental representation of $\mathrm{SU}(N-1)$ and are also charged under $\mathrm{U}(1)$. Notice that the minimum length of such a string is Δ. Since strings have a tension (energy per unit length) $1/(2\pi\alpha')$, this means such strings correspond to fields with a non-zero mass

$$m = \frac{\Delta}{2\pi\alpha'} \, . \tag{5.8}$$

These strings correspond to the "W-bosons" — the fields that have acquired a mass due to the spontaneous breaking of the gauge symmetry.

Now imagine sending the separated brane away by taking $\Delta \to \infty$. The "W-boson" strings become infinitely massive and therefore correspond to very heavy test charges in the $\mathrm{SU}(N-1)$ theory, i.e. Wilson lines. Now if we replace the stack of D3-branes by their $\mathrm{AdS}_5 \times \mathrm{S}^5$ gravitational description, we see that Wilson lines correspond to strings that extend to infinity, i.e. that reach the boundary of AdS. In the classical limit, the expectation value of the Wilson loop operator is then given by the contribution of the string to the saddle-point approximation to the path integral, namely

$$\langle W[C] \rangle = e^{iS^\star_{\mathrm{NG,ren}}[C]} \, , \tag{5.9}$$

where $S^\star_{\mathrm{NG}}[C]$ is the on-shell, holographically renormalised Nambu–Goto action. The string should intersect the boundary of AdS along the curve C defining the Wilson loop. As usual, if we work in Euclidean signature we should replace iS_{NG} with minus the Euclidean action, tot hat the prescription becomes

$$\langle W[C] \rangle = e^{-I_{\mathrm{NG,ren}}} \, . \tag{5.10}$$

We will use the Euclidean version in what follows.

A small caveat. The Wilson loop operators whose expectation values are computed by equation (5.10) are not precisely the operators defined by equation (5.5), but rather a supersymmetric version in which the scalar fields of $\mathcal{N}=4$ SYM appear in the integrand. This operator is sometimes called a Maldacena–Wilson loop. The distinction will not be important for our discussion. From now on, following common practice, we will refer to the supersymmetric versions simply as "Wilson loops".

5.3 Circular Wilson loops in $\mathcal{N} = 4$ SYM

We begin with an example that will allow us to test the prescription for the holographic computation of Wilson loops. Consider $\mathcal{N} = 4$ SYM in in Euclidean signature in flat space $\mathbb{R}^4$. We work in plane polar coordinates (ρ, θ) for two of the spatial directions, so that the metric on $\mathbb{R}^4$ is

$$\mathrm{d}s^2 = \mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\theta^2\,, \tag{5.11}$$

where τ is the Euclidean time. We take the curve C defining our Wilson loop to be a circle of some radius $\rho = \ell$, at constant t and x. Supersymmetry allows the expectation value of this Wilson loop to be calculated at all values of the 't Hooft coupling in $\mathcal{N} = 4$ SYM at large N, with the result

$$\langle W[C]\rangle = \frac{2}{\sqrt{\lambda}} I_1(\sqrt{\lambda}), \tag{5.12}$$

where I_1 is a modified Bessel function. If we expand for large λ, we find that at leading order the logarithm of the Wilson loop is

$$\log\langle W[C]\rangle = \sqrt{\lambda} + \cdots\,, \qquad (\lambda \gg 1)\,. \tag{5.13}$$

We should be able to reproduce this using holography.

To do the holographic calcuation, we adopt coordinates for $\mathrm{AdS}_5 \times \mathrm{S}^5$ in which the metric is

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left(\mathrm{d}\tau^2 + \mathrm{d}x^2 + \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\theta^2 + \mathrm{d}z^2\right) + L^2\,\mathrm{d}\Omega_5^2\,. \tag{5.14}$$

The expectation value of the Wilson loop is determined by the on-shell action of a string that reaches the boundary of AdS_5 at $\rho = \ell$. Let's paramaterise the string with the coordinates $\sigma = (z, \theta)$. The embedding of the string is then specified by how the other directions depend on σ. By translational invariance in the field theory directions we expect that the string will sit at constant values of τ and x, and by rotational invariance on the S^5 we expect it to sit at a constant point on the S^5. Thus the only non-trivial function in the embedding should be $\rho(\sigma)$. Moreover, the rotational symmetry of the Wilson loop suggests that ρ should be independent of σ, so we make the ansatz $\rho = \rho(z)$.

The Euclidean Nambu–Goto action is

$$I_{\mathrm{NG}} = \frac{1}{2\pi\alpha'}\int \mathrm{d}^2\sigma\sqrt{\det\gamma}\,. \tag{5.15}$$

With our ansatz that $\rho = \rho(z)$, the induced metric becomes

$$\gamma_{ab}\,\mathrm{d}\sigma^a\,\mathrm{d}\sigma^b = \frac{L^2}{z^2}\left[\left(1+\rho'^2\right)\mathrm{d}z^2 + \rho^2\,\mathrm{d}\theta^2\right], \tag{5.16}$$

where the prime denotes a derivative with respect to z. Plugging the induced metric into the Nambu–Goto action, we find

$$\begin{aligned} I_{\mathrm{NG}} &= \frac{L^2}{2\pi\alpha'}\int \mathrm{d}z \int_0^{2\pi} \mathrm{d}\theta\,\frac{\rho}{z^2}\sqrt{1+\rho'^2} \\ &= \sqrt{\lambda}\int \mathrm{d}z\,\frac{\rho}{z^2}\sqrt{1+\rho'^2} \end{aligned} \tag{5.17}$$

where in the second line we have performed the integral over θ and used the relation $L^4 = \lambda\alpha'^2$. The equation of motion for $\rho(z)$ which follows from extremising the action has the solution

$$\rho = \sqrt{\ell^2 - z^2}\,, \tag{5.18}$$

which obeys the correct boundary condition $\rho(0) = \ell$. Plugging this solution back into the action, we find the on-shell Nambu–Goto action

$$\begin{aligned} I^\star_{\rm NG} &= \sqrt{\lambda}\,\ell\int_\epsilon^\ell \frac{\mathrm{d}z}{z^2} \\ &= \sqrt{\lambda}\left(\frac{\ell}{\epsilon} - 1\right). \end{aligned} \tag{5.19}$$

The upper limit on the z integral comes from the fact that the solution (5.18) is real only for $z \leq \ell$. The lower limit ϵ is a near-boundary cutoff, which as usual we have introduced because otherwise the integral over z would diverge.

As in previous calculations that we have done, we need to eliminate the ϵ^{-1} divergence with a counterterm. In this case the counterterm we need is the "volume" of the one-dimensional intersection of the string with the cutoff surface at $z = \epsilon$,

$$I_{\rm ct} = c\int_0^{2\pi}\mathrm{d}\theta\,\sqrt{g_{\theta\theta}}|_{z=\epsilon} = 2\pi c L\frac{\ell}{\epsilon}\,, \tag{5.20}$$

for some coefficient c. We see that if we choose $c = -\frac{L}{2\pi\alpha'} = -\frac{\sqrt{\lambda}}{2\pi L}$, this counterterm will exactly cancel the divergence, so that we will obtain a finite renormalised on-shell action for the string,

$$I^\star_{\rm NG,ren} \equiv I^\star_{\rm NG} + I_{\rm ct} = -\sqrt{\lambda}\,. \tag{5.21}$$

Then from the holographic prescription $\langle W\rangle = e^{-I^\star_{\rm NG,ren}}$, we obtain the correct leading-order result at large coupling,

$$\log\langle W[C]\rangle = \sqrt{\lambda}\,. \tag{5.22}$$

5.4 The quark-antiquark potential in $\mathcal{N} = 4$ SYM

Now let's look at rectangular Wilson loops and use the holographic prescription to compute the quark-antiquark potential for strongly-coupled $\mathcal{N} = 4$ SYM in flat space. We should say right away that we can already use conformal symmetry to see that the potential must be of the Coulomb type, and so the theory cannot be confining even at strong coupling. This is because the potential has dimensions of energy, and since $\mathcal{N} = 4$ SYM has no dimensionful parameters, dimensional analysis implies that $V(r) \propto 1/r$. Our aim is to compute the proportionality coefficient.

This time we want to find the on-shell action of a string in the Euclidean $\mathrm{AdS}_5 \times \mathrm{S}^5$ spacetime (5.14) which extends over all τ and touches the boundary at $x = \pm r/2$. We will parameterise the string by (τ, x) and look for the function $z(x)$ the extremises the Nambu–Goto action subject to the boundary condition $z(\pm r/2) = 0$. The induced metric on the string is

$$\gamma_{ab}\,\mathrm{d}\sigma^a\,\mathrm{d}\sigma^b = \frac{L^2}{z^2}\left[\mathrm{d}\tau^2 + (1 + z')^2\,\mathrm{d}x^2\right], \tag{5.23}$$

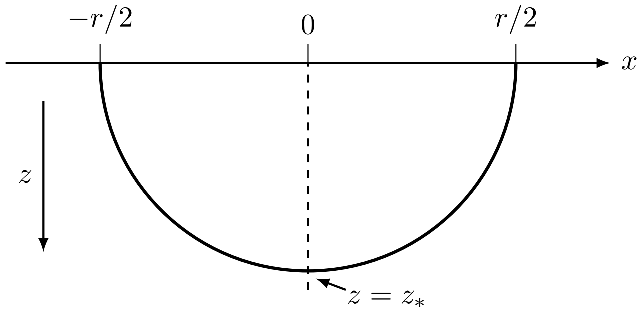


Figure 8: A sketch of the string in AdS_5 dual to the rectangular Wilson loop, at a given Euclidean time τ. The string meets the boundary at $x = \pm r/2$. Its maximal extent into the bulk of AdS is denoted as $z = z_*$ and occurs at $x = 0$.

where the prime denotes differentiation with respect to z. Substituting this into the action (5.15), we find

$$I_{\rm NG} = \frac{TL^2}{2\pi\alpha'} \int_{-r/2}^{r/2} \mathrm{d}x\, \mathcal{I}\,, \qquad \text{where} \qquad \mathcal{I} = \frac{\sqrt{1+z'^2}}{z^2}\,. \tag{5.24}$$

We need to find $z(x)$ that extremises this action. To do this, it is helpful to note that the integrand $\mathcal{I}$ does not depend explicitly on x, and therefore the quantity $h \equiv z'\partial_{z'}\mathcal{I} - \mathcal{I}$ is independent of x on solutions to the equations of motion. Explicitly, we find

$$h = -\frac{1}{z^2\sqrt{1+z'^2}}\,. \tag{5.25}$$

We can write a useful expression for h by noting that the string must have a maximal extent into the bulk of AdS, which by symmetry under the reflection $x \to -x$ we expect to happen at $x = 0$. Defining $z_* = z(0)$ and noting that since z_* is a maximum of $z(x)$ we should have $z'(0) = 0$. See the sketch in figure 8. We can evaluate h at $x = 0$ to find $h = -1/z_*^2$. Equation (5.25) may then be rewritten as

$$z_*^2 = z^2\sqrt{1+z'^2}\,, \qquad \Rightarrow \qquad z'^2 = \frac{z_*^4}{z^4} - 1\,. \tag{5.26}$$

Solve this first-order equation by separation of variables, we find

$$x(z) = \pm z_* \int_{z/z_*}^{1} \mathrm{d}u \frac{u^2}{\sqrt{1-u^4}}\,, \tag{5.27}$$

where the plus or minus sign corresponds to the the fact that there are two values of x for every z. Using the boundary condition that $x(z=0) = \pm r/2$, one finds

$$r = 2\sqrt{\pi}\frac{\Gamma(3/4)}{\Gamma(1/4)} z_*\,, \tag{5.28}$$

where Γ is the Gamma function.

We can use the expression for z' in equation (5.25) to change integration variable in $I_{\rm NG}$ from x to z. The result is

$$I_{\rm NG} = \frac{T\sqrt{\lambda}}{\pi} \int_0^{z_*} \mathrm{d}z \frac{z_*^2}{z^2\sqrt{z_*^4 - z^4}}\,, \tag{5.29}$$

where we have also made the substitution $L^2 = \alpha'\sqrt{\lambda}$. This integral diverges because the integrand blows up as z^{-2} when $z \to 0$. To regularise this we integrate only over $z \geq \epsilon$. Thus, we replace I_{NG} with

$$\begin{aligned} I_{\text{NG}} &= \frac{T\sqrt{\lambda}}{\pi}\int_{\epsilon}^{z_*} \mathrm{d}z \frac{z_*^2}{z^2\sqrt{z_*^4 - z^4}} \,. \\ &= T\sqrt{\lambda}\left[\frac{1}{\pi\epsilon} - \frac{4\pi^2}{\Gamma(1/4)^4}\frac{1}{r}\right] . \end{aligned} \tag{5.30}$$

In the second line we evaluated the integral explicitly and used equation (5.28) to replace z_* with r.

As in previous calculations that we have done, we need to eliminate the ϵ^{-1} divergence with a counterterm. In this case the counterterm we need is the "volume" of the one-dimensional intersection of the string with the cutoff surface at $z = \epsilon$,

$$I_{\text{ct}} = c\int \mathrm{d}\tau \; \sqrt{g_{\tau\tau}}|_{z=\epsilon} = \frac{2cTL}{\epsilon} \,, \tag{5.31}$$

for some coefficient c (the factor of two comes from the fact that the string intersects the boundary at two points). We see that if choose $c = -\sqrt{\lambda}/(2\pi L)$ then this counterterm will precisely cancel the divergence in the Nambu–Goto action, yielding the renormalised action

$$I_{\text{NG,ren}} = I_{\text{NG}} + I_{\text{ct}} = -\frac{4\pi^2\sqrt{\lambda}}{\Gamma(1/4)^4}\frac{T}{r}. \tag{5.32}$$

Recalling that the expectation value of the Wilson loop is given holographically with $\langle W \rangle = e^{-I_{\text{NG,ren}}}$ and that it should be identified with $e^{-TV(r)}$, we can read off the quark-antiquark potential in strongly-coupled $\mathcal{N} = 4$ SYM,

$$V(r) = -\frac{4\pi^2\sqrt{\lambda}}{\Gamma(1/4)^4}\frac{1}{r} \,. \tag{5.33}$$

As anticipated, we find that $V(r) \propto 1/r$ due to conformal symmetry. For comparison, the potential at weak coupling is $V(r) = -\lambda/(4\pi r)$.

5.5 Holographic confinement

The fact that $\mathcal{N} = 4$ SYM is not confining in flat space may seem a bit disappointing, since confinement is such an important property of QCD and it would be nice to be able to study it in a holographic model. Fortunately, there are holographic setups in which one can find area-law behaviour of the rectangular Wilson loop. To do so we need to break conformal symmetry, since conformal field theories always have $V(r) \propto 1/r$. Moreover, to get linear growth of the potential $V(r) \propto r$, we need that the renormalised area of the string grows linearly with r. A typical way that this is realised is to consider models in which there is some kind of "wall" in the bulk of AdS, at some $z = z_0$. Strings corresponding to large-r rectangular Wilson loops will then have a large part of their worldsheet lying flat along the wall, as sketched in figure 9, leading to a renormalised area — and therefore quark-antiquark potential — proportional to r.

As a simple model of this, consider $\mathcal{N} = 4$ SYM in flat space, and compactify one of the spatial directions on a circle of circumference ℓ. In the deep IR, at length scales much

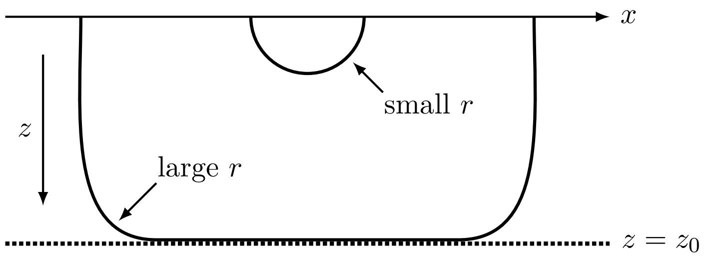


Figure 9: Strings in a geometry with a "wall" at some $z = z_0$. Strings corresponding to small-separation rectangular Wilson loops behave similarly to in AdS without the wall. Strings corresponding to large separation r have a significant section of their worldsheet that lies along the wall, leading to a renormalised area proportional to r.

larger than ℓ, physics is effectively $(2+1)$-dimensional. The bulk five-dimensional metric is dual to this setip is, in Euclidean signature,

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left(\mathrm{d}\tau^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + u(z)\ell^2\,\mathrm{d}\theta^2 + \frac{\mathrm{d}z^2}{u(z)}\right), \qquad u(z) = 1 - \frac{z^4}{z_0^4}, \tag{5.34}$$

where we have paramaterised the circle with an angle θ that has periodicity 2π. Notice that this is the same as the AdS-Schwarzschild solution (4.6), except that the compactified direction is a spatial direction rather than the Euclidean time. The same kind of calculation that we do to compute the Hawking temperature for black holes shows that to avoid a singularity, z_0 must be related to ℓ by

$$z_0 = 2\ell\,. \tag{5.35}$$

We make the same ansatz for the string as we made in the previous section, which leads to the Nambu–Goto action

$$I_{\mathrm{NG}} = \frac{TL^2}{2\pi\alpha'}\int_{-r/2}^{r/2}\mathrm{d}x\,\mathcal{I}\,, \qquad \text{where} \qquad \mathcal{I} = \frac{1}{z^2}\sqrt{1+\frac{z'^2}{u(z)}}\,. \tag{5.36}$$

Just like before, the integrand $\mathcal{I}$ is independent of x, so that we have a conserved quantity $z'\partial_{z'}\mathcal{I} - \mathcal{I}$. Writing this in terms of the maximal extent z_* of the string into the bulk, we find

$$z_*^2 = z^2\sqrt{1+\frac{z'^2}{u(z)}} \qquad \Rightarrow \qquad z' = \pm\sqrt{u(z)}\sqrt{\frac{z_*^4}{z^4}-1}\,. \tag{5.37}$$

We can integrate this to find

$$\begin{aligned} r &= 2\sqrt{\pi}\frac{\Gamma(3/4)}{\Gamma(1/4)}\,{}_2F_1\left(\frac{1}{2},\frac{3}{4};\frac{5}{4};\frac{z_*^4}{z_0^4}\right) \\ &\approx \frac{z_0}{4}\left[2\log\left(\frac{8z_0}{z_0-z_*}\right)-\pi\right], \qquad (\text{at } z_* \approx z_0)\,. \end{aligned} \tag{5.38}$$

where ${}_2F_1$ is the hypergeometric function. In the second line we have expanded the expression z_* close to z_0. We see that r diverges in the limit that $z_* \to z_0$, consistent with our expectation that for large r the string will approach the wall at $z = z_0$. Inverting the approximate expression for r, we find that at $r \gg \ell$, the maximum extent of the string into the bulk is

$$z_* \approx z_0 \left[1 - 8\exp\left(-\frac{r}{\ell} - \frac{\pi}{2}\right)\right]. \tag{5.39}$$

Using the expression for z' in equation (5.37) to change integration variables from x to z, the Nambu–Goto action becomes

$$I_{\text{NG}} = \frac{TL^2}{\pi\alpha'} \int_\epsilon^{z_*} \mathrm{d}z \frac{z_0^2 z_*^2}{z^2\sqrt{(z_0^4 - z^4)(z_*^4 - z^4)}}\,, \tag{5.40}$$

where as usual we have introduced a small-z cutoff ϵ. Evaluating this integral gives

$$I_{\text{NG}} = \frac{L^2 T}{2\pi\alpha'}\left[\frac{2}{\epsilon} + \frac{18\sqrt{\pi}}{5}\frac{z_*^3}{z_0^4}\frac{\Gamma(3/4)}{\Gamma(1/4)}\,{}_2F_1\left(\frac{1}{2},\frac{7}{4};\frac{9}{4};\frac{z_*^4}{z_0^4}\right) - \frac{z_0^4 + z_*^4}{z_0^5 z_*^2}\right] \tag{5.41}$$

The counterterm in equation (5.31) cancels off the $1/\epsilon$ divergence, as before, so that we are left with a finite renormalised action

$$I_{\text{NG,ren}} = \frac{L^2 T}{2\pi\alpha'}\left[\frac{18\sqrt{\pi}}{5}\frac{z_*^3}{z_0^4}\frac{\Gamma(3/4)}{\Gamma(1/4)}\,{}_2F_1\left(\frac{1}{2},\frac{7}{4};\frac{9}{4};\frac{z_*^4}{z_0^4}\right) - \frac{z_0^4 + z_*^4}{z_0^5 z_*^2}\right] \tag{5.42}$$

We can then read off the quark-antiquark potential $V(r) = I_{\text{NG,ren}}/T$. Expanding for $z_* \approx z_0$ and using equation (5.39) to express z_* with ℓ, one finds that the leading-order behaviour of the potential at large r is

$$V(r) \approx \frac{3L^2 r}{8\pi\alpha'\ell^2}\,. \tag{5.43}$$

So we do indeed expect confinement in the IR theory.

6 Entanglement entropy

6.1 Entanglement entropy in quantum systems

A popular topic of research in holography is a quantity called *entanglement entropy*. This is something that can be defined in any quantum system which has a Hilbert space $\mathcal{H}$ which factorises into a tensor product of two subspaces $\mathcal{A}$ and $\mathcal{A}^c$ (the c stands for "complement"),

$$\mathcal{H} = \mathcal{A} \otimes \mathcal{A}^c\,. \tag{6.1}$$

Given the density matrix ρ that specifies the state of the system in $\mathcal{H}$, we can define the *reduced density matrix* $\rho_\mathcal{A}$ for $\mathcal{A}$ be tracing out states in $\mathcal{A}^c$,

$$\rho_\mathcal{A} = \operatorname{tr}_{\mathcal{A}^c}\rho \equiv \sum_{|e\rangle}\langle e|\rho|e\rangle\,, \tag{6.2}$$

where the sum on the right is over a complete basis of states for $\mathcal{A}^c$. The reduced density matrix is an operator on $\mathcal{A}$. The entanglement entropy $S_\mathcal{A}$ of $\mathcal{A}$ with its complement is defined as the von Neumann entropy of the reduced density matrix,

$$S_\mathcal{A} = -\operatorname{tr}_\mathcal{A}(\rho_\mathcal{A}\log\rho_\mathcal{A})\,. \tag{6.3}$$

When the original density matrix ρ is that of a pure state, the entanglement entropy provides a measure of the quantum entanglement between $\mathcal{A}$ and $\mathcal{A}^c$.[8] For example suppose $\mathcal{A}$ and $\mathcal{A}^c$ are two qubits. We denote the basis states in $\mathcal{A}$ as $|0\rangle$ and $|1\rangle$, and the basis states in $\mathcal{A}^c$ as $|-\rangle$ and $|+\rangle$. The full Hilbert space $\mathcal{H}$ is then the space spanned by the product states $\{|0\rangle|-\rangle\,, |0\rangle|+\rangle\,, |1\rangle|-\rangle\,, |1\rangle|+\rangle\,\}$. Let's consider the family of pure states with density matrix

$$\rho = |\psi\rangle\langle\psi|\,, \qquad |\psi\rangle = \cos\theta|0\rangle|+\rangle + \sin\theta|1\rangle|-\rangle\,, \tag{6.4}$$

for some θ. If we trace over the states in $\mathcal{A}^c$ we find

$$\rho_{\mathcal{A}} = \cos^2\theta|0\rangle\langle 0| + \sin^2\theta|1\rangle\langle 1|\,. \tag{6.5}$$

This is a diagonal matrix in the $\{|0\rangle, |1\rangle\}$ basis, so it is straightforward to compute the von Neumann entropy,

$$S_{\mathcal{A}} = -\cos^2\theta\log\Big(\cos^2\theta\Big) - \sin^2\theta\log\Big(\sin^2\theta\Big)\,. \tag{6.6}$$

This vanishes when θ is an integer multiple of $\pi/2$, in which case $|\psi\rangle$ in equation (6.4) is a product state. The entanglement entropy is maximal for $\theta = \pi/4$ (plus integer multiples of $\pi/2$), when $|\cos\theta| = |\sin\theta| = 1/\sqrt{2}$. This is the case that is usually called maximally entangled.

In QFT, we usually take the $\mathcal{A}$ to be the set of states for a region of space at some fixed time. We will also use $\mathcal{A}$ to denote this region. The entanglement entropy $S_{\mathcal{A}}$ then measures the entanglement between that region and its surroundings. This is typically UV divergent, due to short range entanglement between modes on either side of the boundary $\partial\mathcal{A}$ of $\mathcal{A}$. If we regularise this divergence by introducing a short-distance UV cutoff ϵ, one then finds

$$S_{\mathcal{A}} \sim \frac{\text{area}[\partial\mathcal{A}]}{\epsilon^{d-2}}\,. \tag{6.7}$$

In general there can be subleading divergences as well, the structure of which depends on dimension. Concretely, in d spacetime dimensions and for $\mathcal{A}$ with characteristic length scale ℓ, one finds

$$S_{\mathcal{A}} = k_{d-2}\frac{\ell^{d-2}}{\epsilon^{d-2}} + k_{d-4}\frac{\ell^{d-4}}{\epsilon^{d-4}} + \cdots + \begin{cases} k_0\,, & d \text{ odd,} \\ k_0\log(\ell/\epsilon) + \tilde{k}_0\,, & d \text{ even,} \end{cases} \tag{6.8}$$

for some coefficients k_{d-2}, k_{d-4}, etc. Most of the coefficients depend on how you regularise the UV divergence in the entanglement entropy. For instance if you rescale the cutoff $\epsilon \to \alpha\epsilon$ for some constant α, all of the coefficients will change by factors of α except for the coefficient that we have called k_0. The term involving k_0 is therefore often called the *universal* part of the entanglement entropy, and it is this term that contains the most interesting physical information.

For example, consider a 2D CFT in flat Minkowski space. A constant time slice is then just the real line $\mathbb{R}$. We take the region $\mathcal{A}$ to be an interval along this line of length 2ℓ. A famous result of 2D CFT is that the entanglement entropy of such an interval is

$$S_{\mathcal{A}} = \frac{c}{3}\log(\ell/\epsilon) + \tilde{k}_0\,, \tag{6.9}$$

[8]When ρ describes a mixed state the entanglement entropy is contaminated by the entropy of ρ. For instance, if ρ is a thermal state and $\mathcal{A}$ is large compared to $\mathcal{A}^c$, the entanglement entropy is approximately equal to the thermal entropy, which has nothing to do with entanglement.

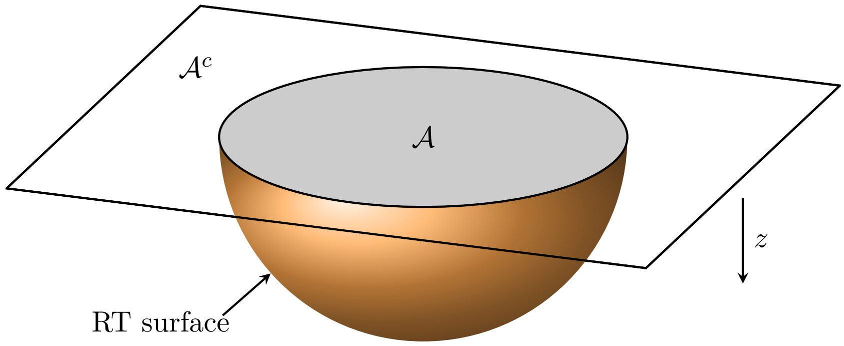


Figure 10: A sketch of a constant time slice in AdS. The parallelogram the top represents the boundary, which has been partitioned into a subregion $\mathcal{A}$ and its complement $\mathcal{A}^c$. Extending below the boundary is the bulk. The entanglement entropy is proportional to the area of the RT surface, the surface of minimal area homologous to $\mathcal{A}$.

where c is the central charge of the CFT, while $\tilde{k}_0$ is an arbitrary constant that depends on how you regularise the UV divergence. For a 4D CFT in flat space, if we take $\mathcal{A}$ to be a ball of radius ℓ, the entanglement entropy takes the form

$$S_{\mathcal{A}} = k_2 \frac{\ell^2}{\epsilon^2} - 4a \log(\ell/\epsilon) + \tilde{k}_0 \,. \tag{6.10}$$

where a is the type A Weyl anomaly coefficient we encountered in section 4.4.

6.2 Holographic entanglement entropy

Entanglement entropy in a holographic QFT may be computed from its gravitational dual via the Ryu–Takayanagi (RT) prescription, which works as follows. First, thinking of the QFT as living at the boundary of the (asymptotically) AdS spacetime, the subregion $\mathcal{A}$ becomes a subregion of the boundary of a constant-time slice of AdS. To find the entanglement entropy, we need to find the surface γ of minimal area that is homologous to $\mathcal{A}$. The RT formula for the entanglement entropy is then

$$S_{\mathcal{A}} = \frac{\text{area}[\gamma]}{4G} \,. \tag{6.11}$$

The surface γ is usually called the RT surface. The homology requirement means that there is a subregion $\mathcal{R}$ of the the constant time slice whose boundary is the union of $\mathcal{A}$ and γ, i.e. $\partial\mathcal{R} = \mathcal{A} \cup \gamma$. Usually, this means that γ meets the AdS boundary at $\partial\mathcal{A}$, as depicted in figure 10.

Let's exhibit this formula with an example. We consider AdS$_{d+1}$ in Poincaré coordinates, using spherical polar coordinates on the boundary, so that the metric is

$$\mathrm{d}s^2 = \frac{L^2}{z^2} \left(-\,\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\,\mathrm{d}\Omega_{d-2}^2 + \mathrm{d}z^2 \right). \tag{6.12}$$

We take $\mathcal{A}$ to be the ball $r \leq \ell$ from some radius ℓ. Working at constant time t, consider a $(d-1)$-dimensional surface γ parameterised by coordinates $\vec{\sigma}$ which we take to be r

and the coordinates on the S^{d-2}. The embedding of such a surface is specified by how z depends on $\vec{\sigma}$. Since our choice of $\mathcal{A}$ preserves the rotational symmetry on the boundary, we expect that the minimal surface should too, so that $z = z(r)$. The area of such a surface is

$$\mathrm{area}[\gamma] = L^{d-1}\mathrm{vol}(\mathrm{S}^{d-2}) \int \mathrm{d}r \frac{r^{d-2}}{z^{d-1}} \sqrt{1+z'^2}\,, \tag{6.13}$$

where the prime denotes differentiation with respect to r, and $\mathrm{vol}(\mathrm{S}^{d-2}) = 2\pi^{(d-1)}/\Gamma(\frac{d-1}{2})$ is the volume of a unit $(d-2)$-sphere.

We want to minimise this area subject to the boundary condition $z(\ell) = 0$ (this is the homology constraint). The Euler–Lagrange equation for $z(r)$ is

$$\frac{rzz''}{1+z'^2} + (d-2)zz' + (d-1)r = 0\,, \tag{6.14}$$

which is solved by

$$z = \sqrt{\ell^2 - r^2}\,. \tag{6.15}$$

In other words, the RT surface in this example is just a hemisphere in the (z, r) coordinates, of coordinate radius ℓ. Plugging this solution back into equation (6.13) for the area and using the RT formula (6.11) we find

$$S_{\mathcal{A}} = \frac{\pi^{(d-1)/2}}{2\Gamma(\frac{d-1}{2})G} \ell L^{d-1} \int_0^\ell \mathrm{d}r \frac{r^{d-2}}{(\ell^2 - r^2)^{d/2}}\,, \tag{6.16}$$

where we have also substituted the formula for the volume of the sphere. This integral diverges due to the behaviour of the integrand as $r \to \ell$, which from equation (6.15) corresponds to $z \to 0$. This is the expected UV divergence of the entanglement entropy in QFT. To regularise this divergence we impose a UV cutoff at $z = \epsilon$, meaning that we only integrate over $r \leq \sqrt{\ell^2 - \epsilon^2}$.

The regularised integral may be performed explicitly for arbitrary d in terms of a hypergeometric function, but the result is not so enlightening. Instead we will look at several small values of ϵ and in each case expand the entanglement entropy for small ϵ. First, for $d = 2$ we find

$$S_{\mathcal{A}} = \frac{L}{2G}\left[\log(\ell/\epsilon) - \log 2\right]\,, \qquad (d = 2)\,. \tag{6.17}$$

Comparing to equation (6.9), we see that the entanglement entropy takes the expected form, and we can read off that the dual CFT has central charge $c = 3L/2G$. Next, for $d = 3$ we find

$$S_{\mathcal{A}} = \frac{\pi L^2}{2G}\left(\frac{\ell}{\epsilon} - 1\right)\,, \qquad (d = 3)\,. \tag{6.18}$$

Again this has the expected structure in terms of powers of ϵ. We read off the universal term as $k_0 = -\pi L^2/(2G)$. Finally, for $d = 4$ we find

$$S_{\mathcal{A}} = \frac{\pi L^3}{2G}\left[\frac{\ell^2}{\epsilon^2} - \log(\ell/\epsilon) - \log 2 - \frac{1}{2}\right]\,, \qquad (d = 4)\,. \tag{6.19}$$

Comparing to equation (6.10) we again see that this matches the expected divergence structure, and we can read off that the Weyl anomaly coefficient is $a = \pi L^3/8G$. Using equation (4.4) to express Newton's constant in terms of N^2, for $\mathcal{N} = 4$ SYM this corresponds to $a = N^2/4$, the same as we found from the Casimir energy in equation (4.53).

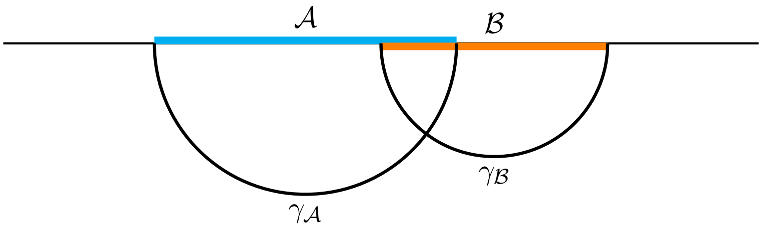
6.3 Strong subadditivity

As a check of the Ryu–Takayanagi prescription, we will check that entanglement entropies derived from it satisfy an inequality that should hold for any quantum system. Consider a Hilbert space which is a tensor product of three factors, $\mathcal{H} = \mathcal{A} \otimes \mathcal{B} \otimes \mathcal{C}$. The sum of the entanglement entropies of $\mathcal{A}$ and $\mathcal{B}$ with their complements satisfies an inequality called strong subadditivity,

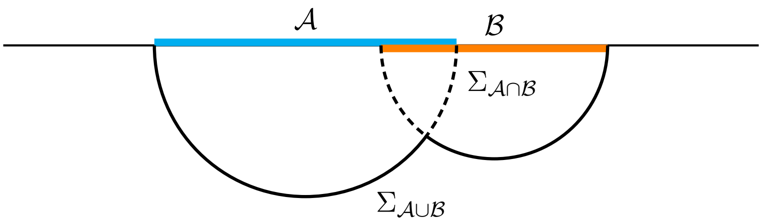
$$S_{\mathcal{A}} + S_{\mathcal{B}} \geq S_{\mathcal{A}\cup\mathcal{B}} + S_{\mathcal{A}\cap\mathcal{B}} \,. \tag{6.20}$$

Holographic entanglement entropy derived from the RT formula obeys this inequality, a fact which has that following beautiful geometric proof.

Below is a sketch of a time slice of an asymptotically AdS spacetime. The line at the top is the boundary, and we have partitioned it into regions $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$. Below the line is the bulk of AdS and we have sketched the RT surfaces $\gamma_{\mathcal{A}}$ and $\gamma_{\mathcal{B}}$ used for computing the entanglement entropies on the left-hand side of the strong subadditivity inequality (6.20).



We can chop up $\gamma_{\mathcal{A}}$ and $\gamma_{\mathcal{B}}$ and glue the various pieces together to get bulk surfaces $\Sigma_{\mathcal{A}\cup\mathcal{B}}$ and $\Sigma_{\mathcal{A}\cap\mathcal{B}}$, homologous to $\mathcal{A}\cup\mathcal{B}$ and $\mathcal{A}\cap\mathcal{B}$ respectively:



Since RT surface for $\mathcal{A}\cup\mathcal{B}$, which we denote $\gamma_{\mathcal{A}\cup\mathcal{B}}$, is the surface of minimal area homologous to $\mathcal{A}\cup\mathcal{B}$, we must have that area$[\gamma_{\mathcal{A}\cup\mathcal{B}}] \leq$ area$[\Sigma_{\mathcal{A}\cup\mathcal{B}}]$. The same logic tells us that the RT surface for $\mathcal{A}\cap\mathcal{B}$ satisfies area$[\gamma_{\mathcal{A}\cap\mathcal{B}}] \leq$ area$[\Sigma_{\mathcal{A}\cap\mathcal{B}}]$. We therefore find

$$\begin{aligned} \text{area}[\gamma_{\mathcal{A}}] + \text{area}[\gamma_{\mathcal{B}}] &= \text{area}[\Sigma_{\mathcal{A}\cup\mathcal{B}}] + \text{area}[\Sigma_{\mathcal{A}\cap\mathcal{B}}] \\ &\geq \text{area}[\gamma_{\mathcal{A}\cup\mathcal{B}}] + \text{area}[\gamma_{\mathcal{A}\cap\mathcal{B}}] \,. \end{aligned} \tag{6.21}$$

Using the RT formula (6.11) this becomes the strong subadditivity inequality (6.20).[9]

References

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[9] In the pictures we drew above we assumed that the intersection $\mathcal{A}\cap\mathcal{B}$ was non-empty. The proof still works if $\mathcal{A}$ and $\mathcal{B}$ do not overlap. I encourage you to think about how.

[2] Jens O. Andersen, Qianqian Du, Michael Strickland, and Ubaid Tantary. “N=4 supersymmetric Yang-Mills thermodynamics from effective field theory”. In: *Phys. Rev. D* 105.1 (2022), p. 015006. DOI: 10.1103/PhysRevD.105.015006. arXiv: 2111.12160 [hep-th].

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